

Intensity Transformation

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Image enhancement

Image enhancement is the procedure of **improving the quality and information content** of original data before processing. Common practices include contrast enhancement, spatial filtering etc.

Domain of image processing

- ❑ Spatial Domain (Direct operation on image)
- ❑ Frequency domain (Operation after Fourier transformation)

Spatial processing

$$g(x,y) = T [f(x,y)]$$

□ $f(x,y)$ □ Input Image

□ $g(x,y)$ □ Output image

□ T □ Operation/ Transform method

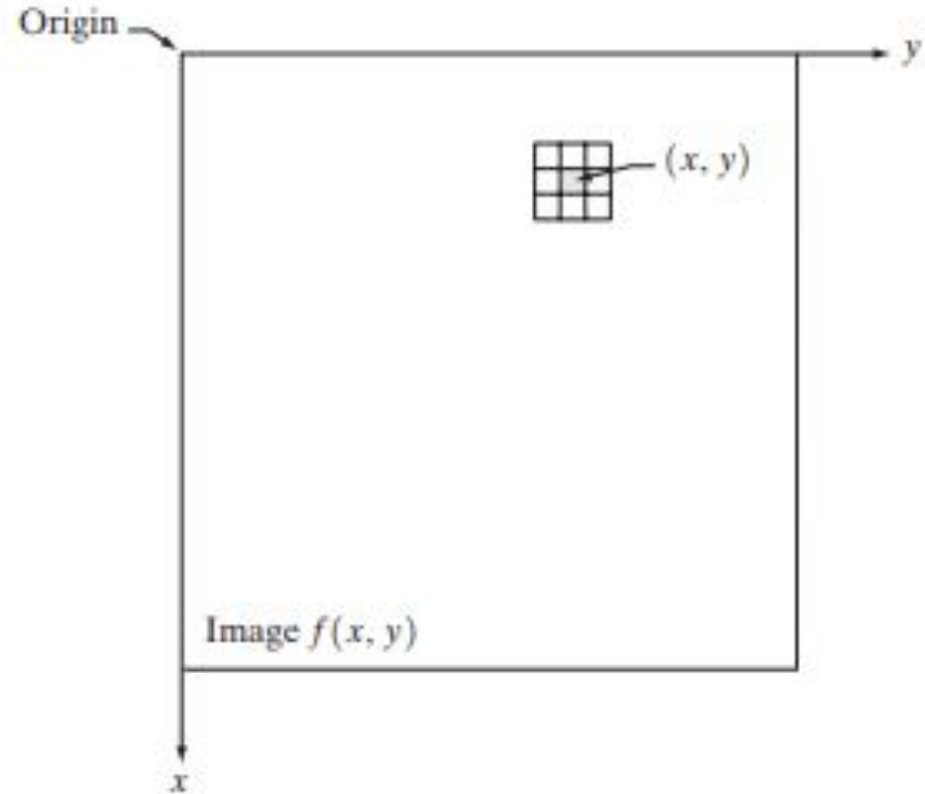
Spatial processing

$$g(x,y) = T [f(x,y)]$$

□ $f(x,y)$ □ Input Image

□ $g(x,y)$ □ Output image

□ T □ Operation/ Transform
method defined over the
neighbourhood of point (x,y)



Spatial processing

For smallest neighbourhood of size $1 * 1$, g depends only on value of f at a single point (x,y) and T becomes a intensity transformer function. Also called gray level mapping.

Hence for simplicity of notation it is described by

$$s = T(r)$$

Here s and r denotes intensity of g and f at a point (x,y)

Spatial processing

- Intensity transformation

 - (Operates on single pixel)

- Spatial Filtering

 - (Operates in neighbourhood)

Intensity Transformation

Intensity transformations are applied on images for contrast manipulation or image thresholding. These are in the spatial domain, i.e. they are performed directly on the pixels.

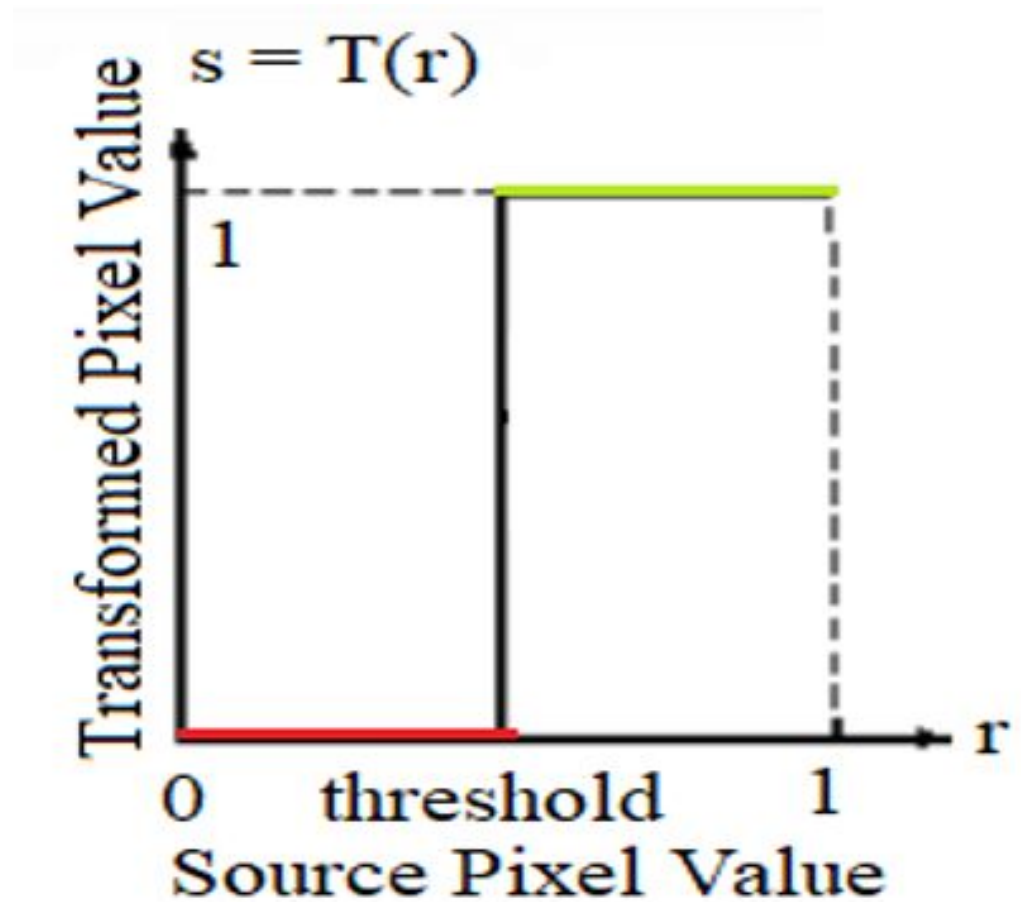
Simplest off all image processing techniques.

Intensity Transformation

Thresholding:

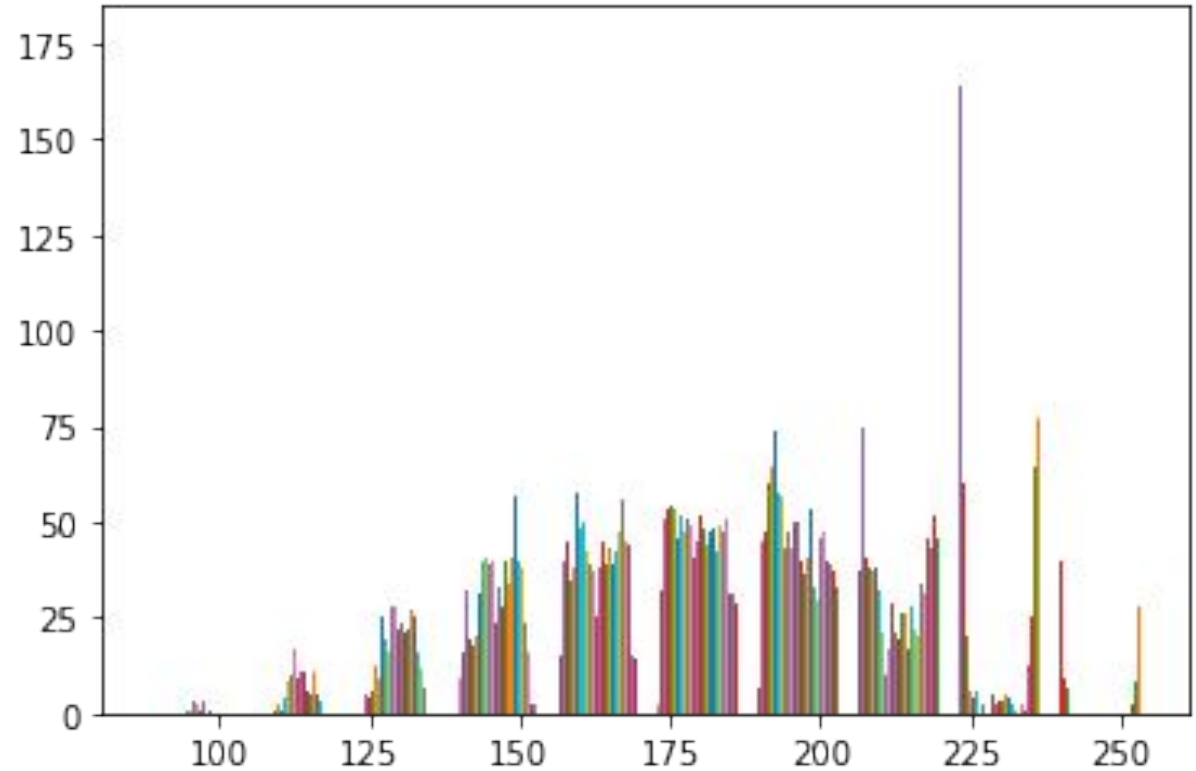
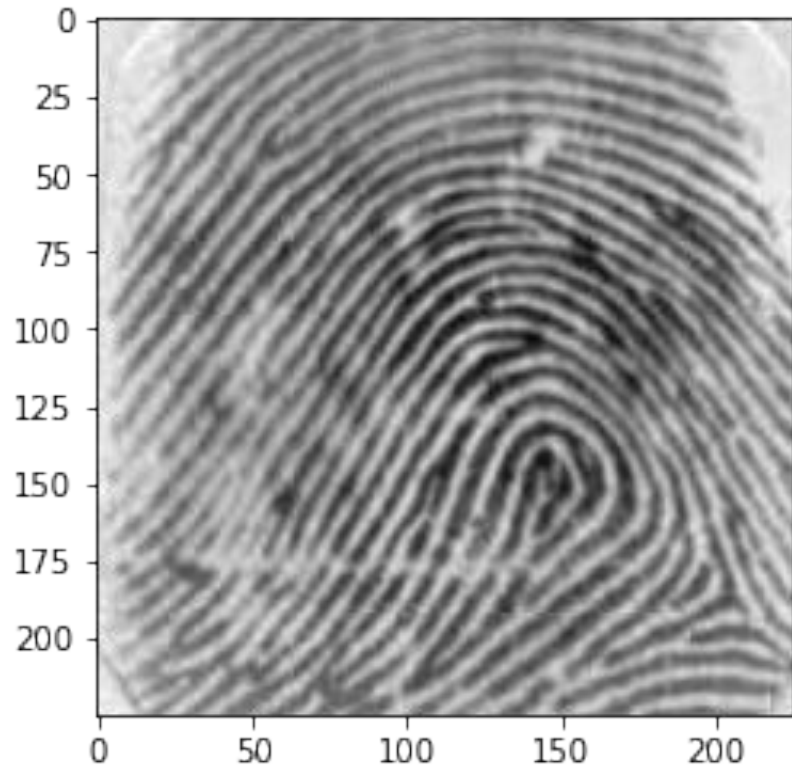
$s = 0$ if $r > \text{Threshold}$

$s = 255$ if $r \leq \text{Threshold}$



Intensity Transformation

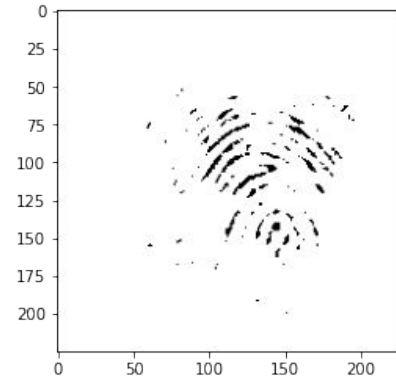
Thresholding



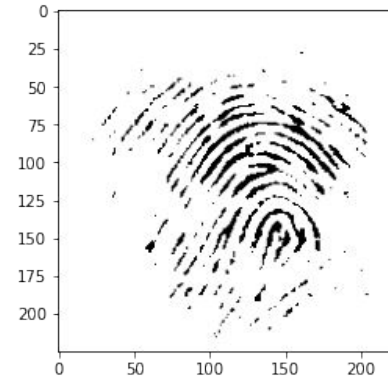
Intensity Transformation

Thresholding

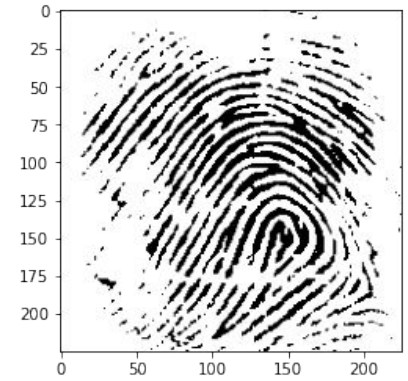
$T = 120$



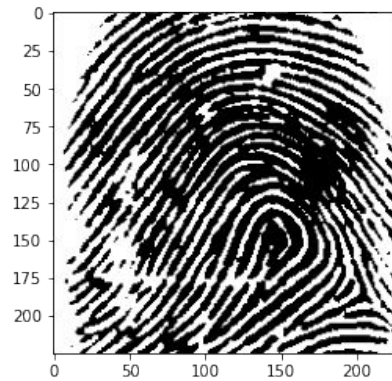
$T = 135$



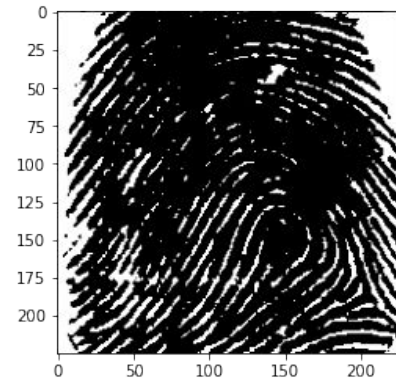
$T = 155$



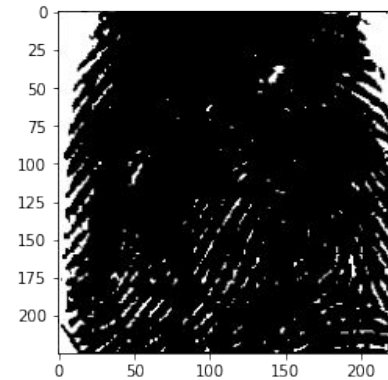
$T = 187$



$T = 205$

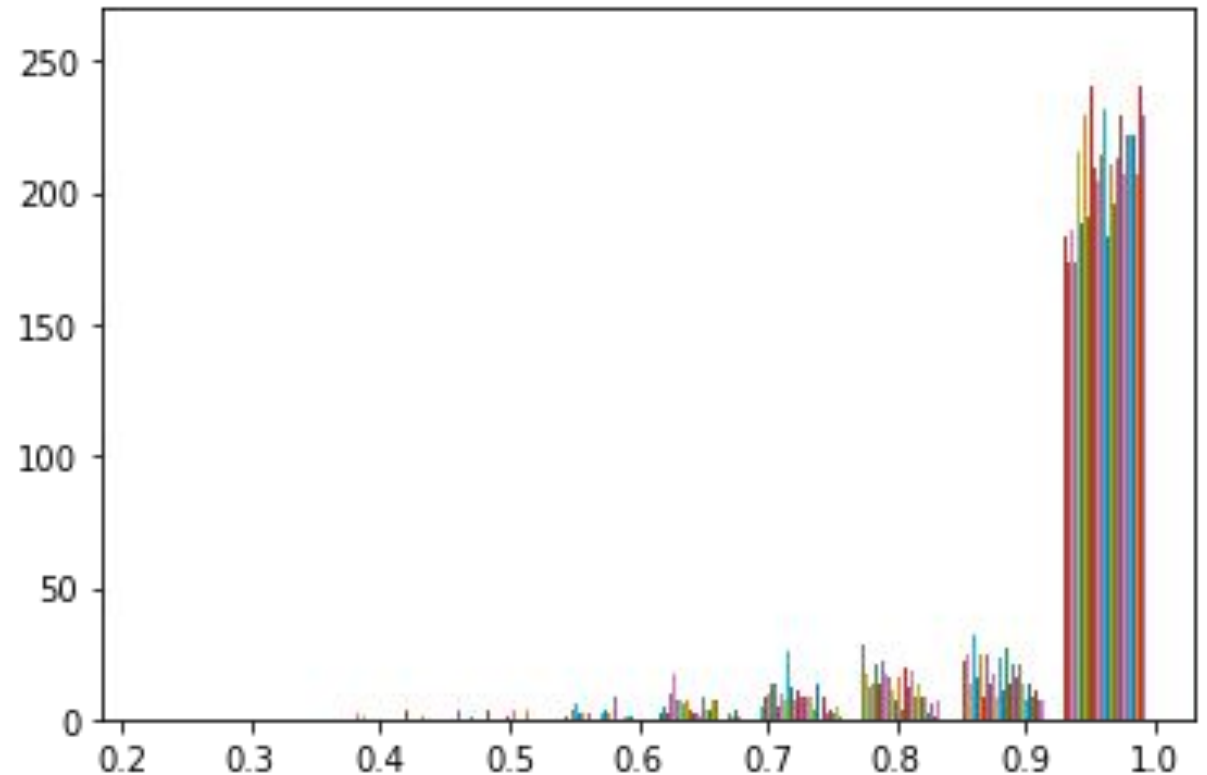
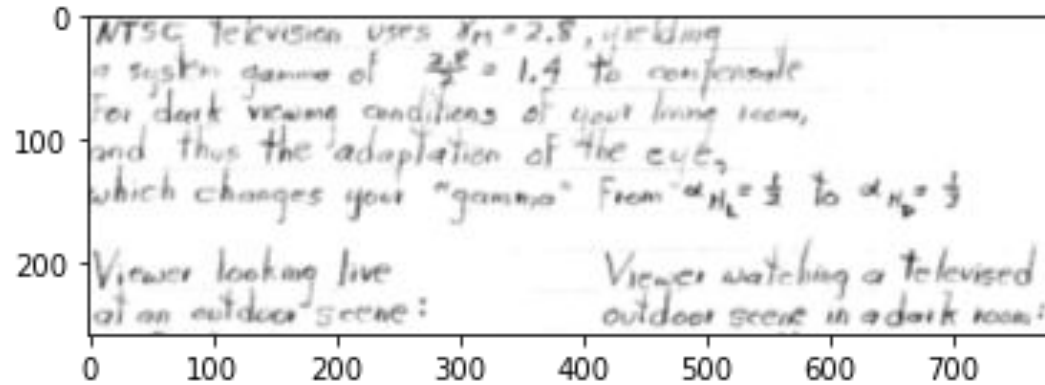


$T = 220$



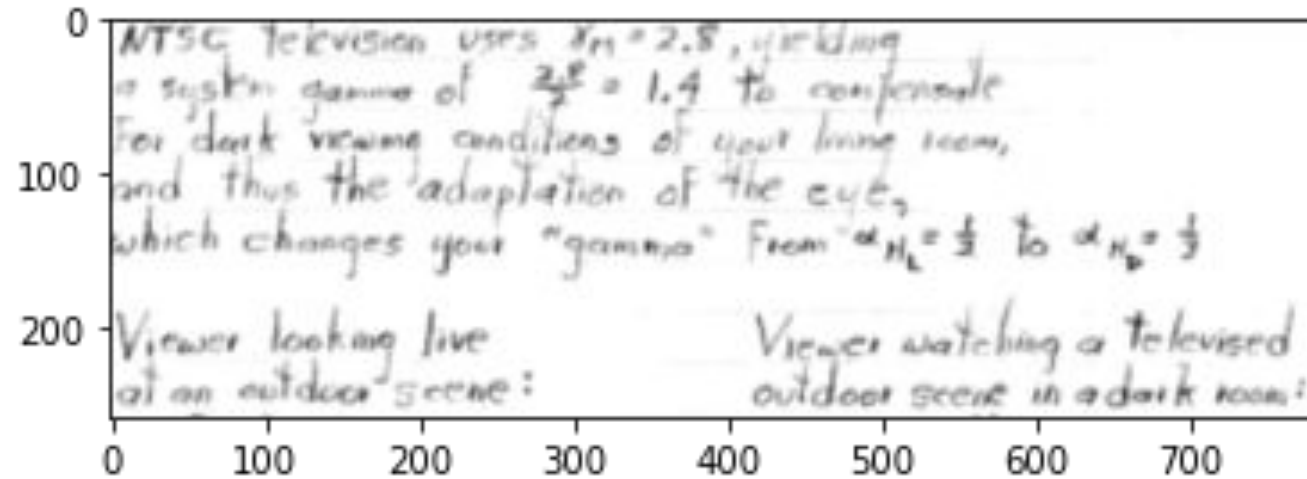
Intensity Transformation

Thresholding

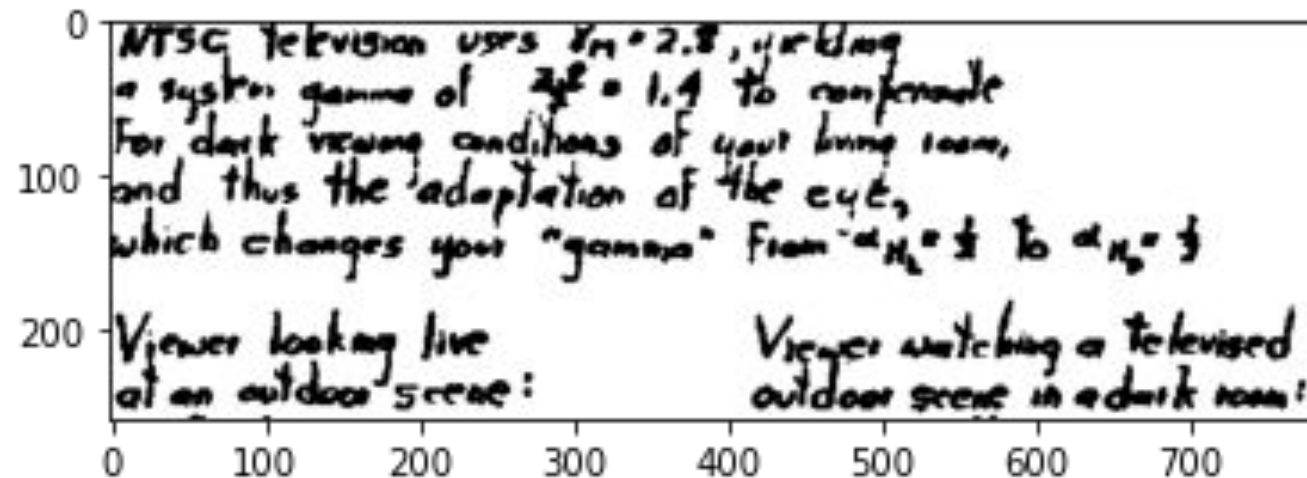


Intensity Transformation

Thresholding



T = 0.92



Intensity Transformation

Some basic subjective quality measurement terminologies:

1. Brightness
2. Contrast
3. Sharpness

Intensity Transformation

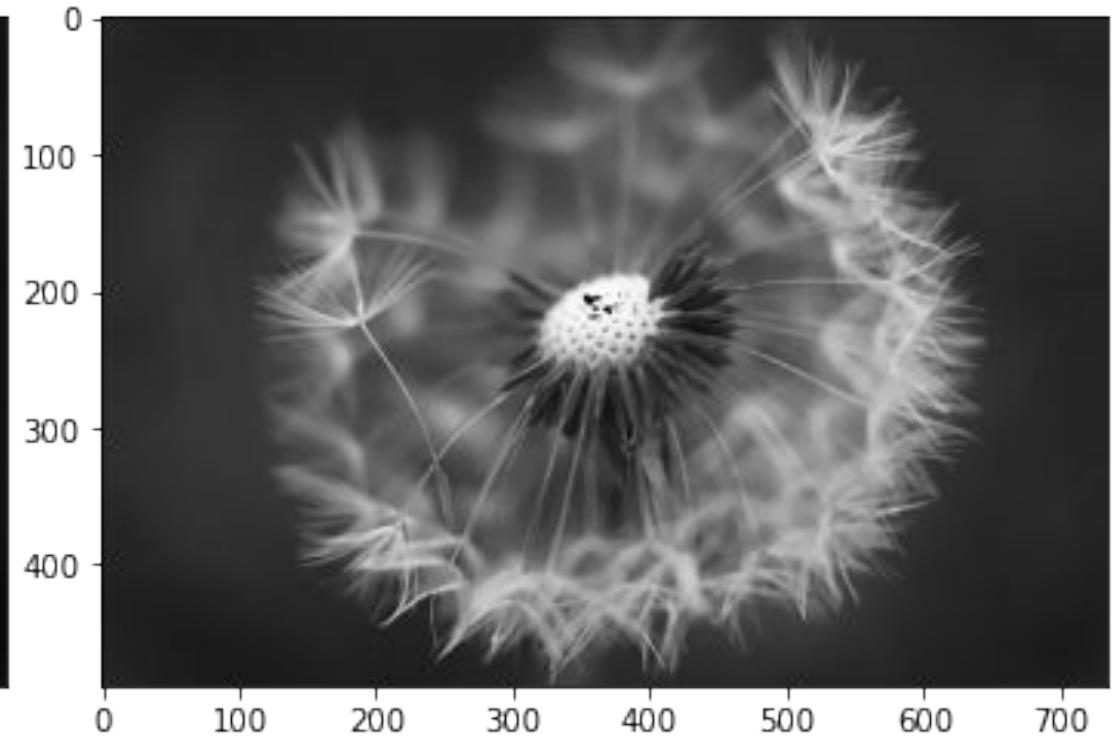
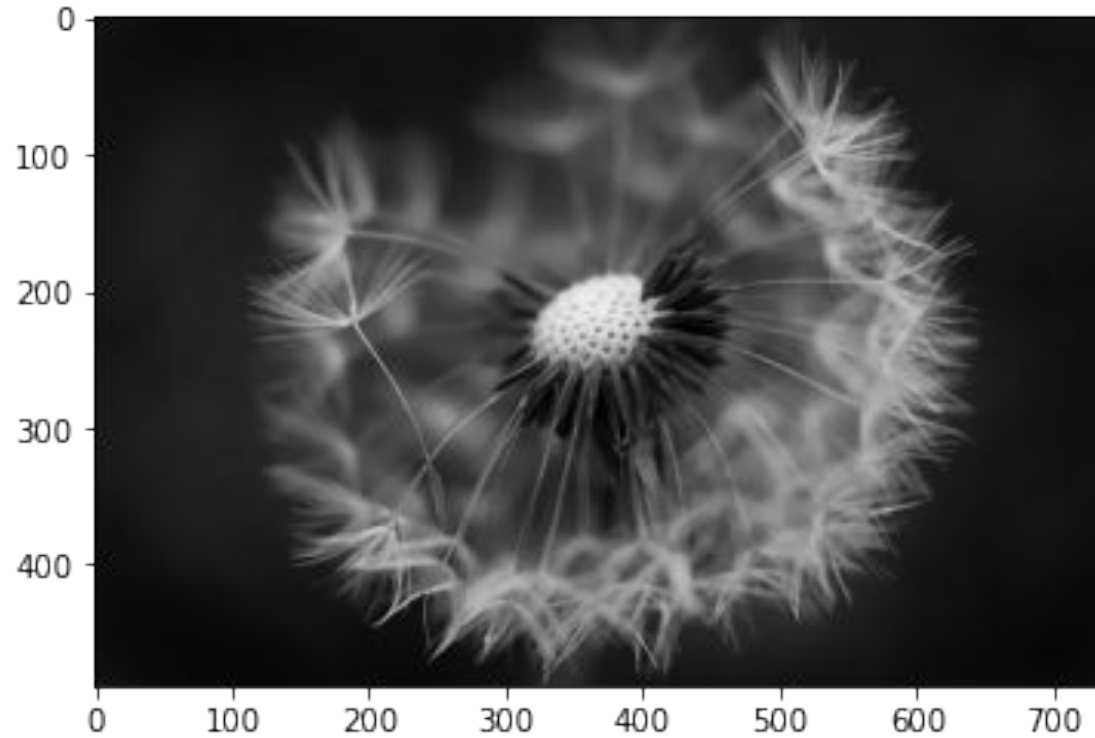
Brightness: In terms of digital image processing, brightness is more properly described as the measured intensity of all the pixels comprising an ensemble that constitutes the digital image.

Traditional brightness simply brightness the entire image from the **shadows to the highlights equally**.

Intensity Transformation

Brightness

$\text{img}+20$

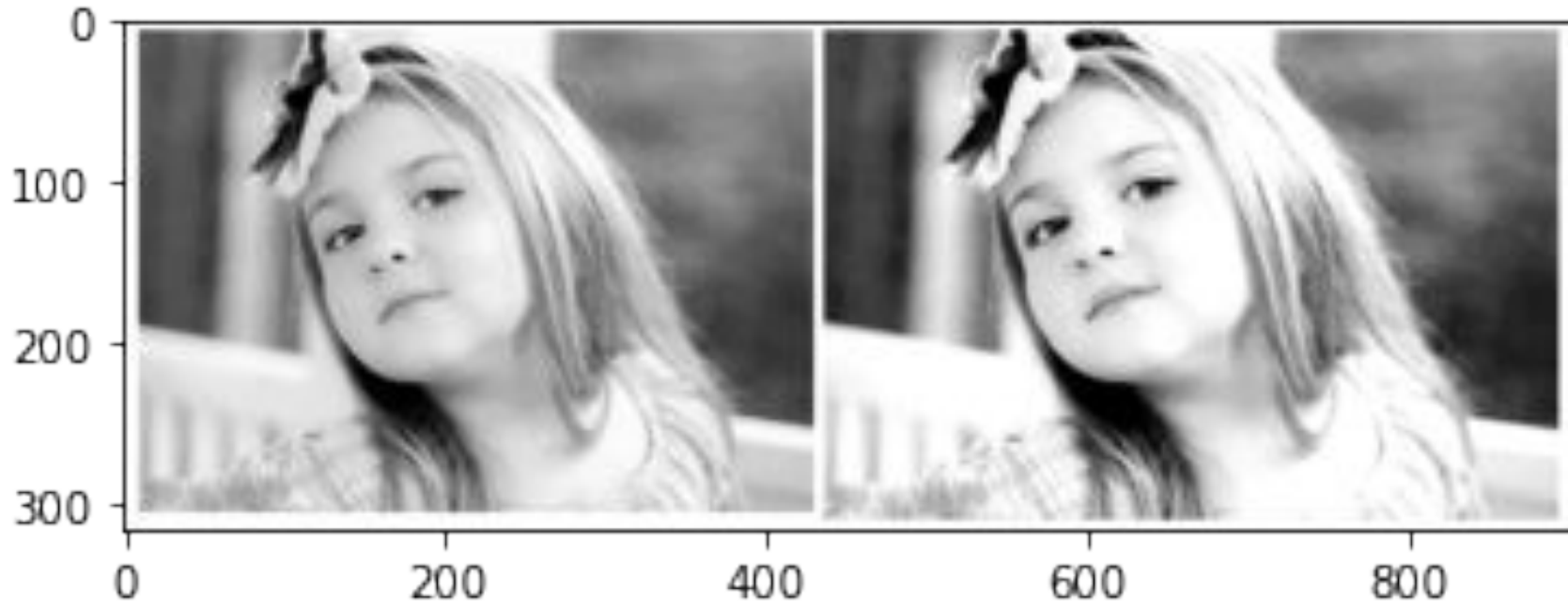


Intensity Transformation

Contrast: Contrast makes an object **distinguishable**. It is the **separation between** the **darkest** and **brightest** areas of the image. **Increase contrast** and you increase the separation between dark and bright, making shadows darker and highlights brighter.

Intensity Transformation

Contrast



Low contrast

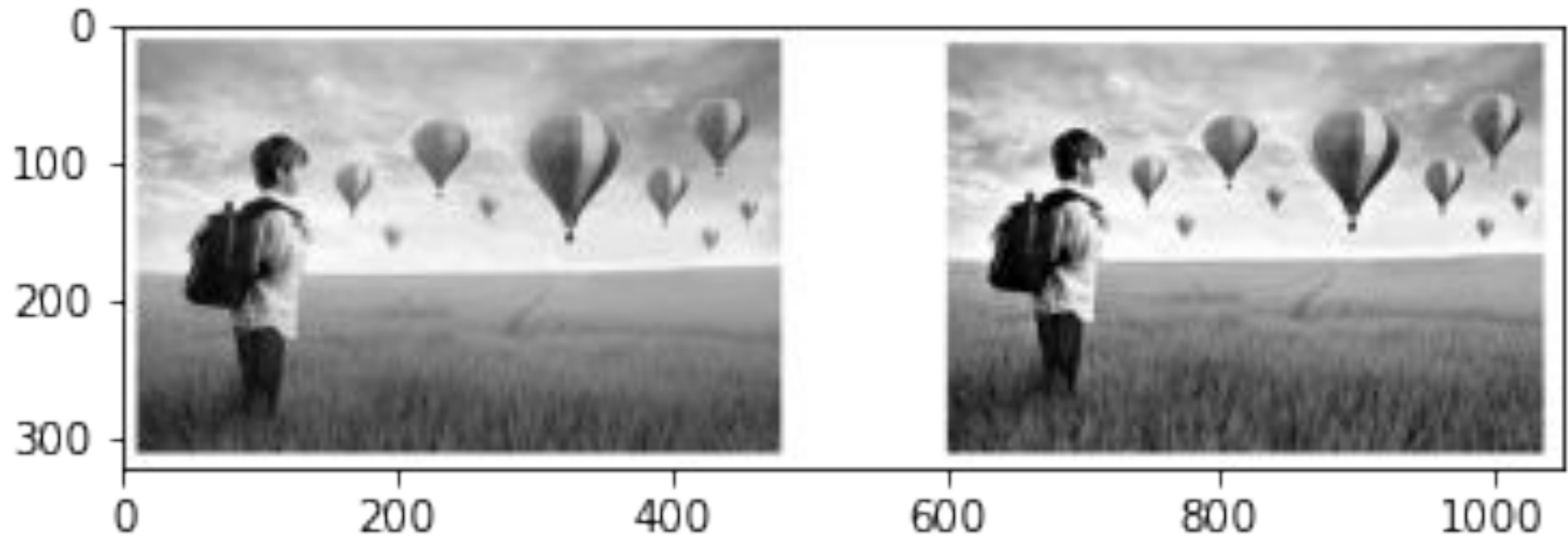
High contrast

Intensity Transformation

Sharpness: Sharpness can be defined as **edge contrast** , that is, the contrast along edges in a photo. When we increase sharpness, we increase the contrast only along/near edges in the photo while leaving smooth areas of the image alone.

Intensity Transformation

Sharpness



Some basic Intensity transformation Functions

There are three types of transformation:

- 1.Linear

- 2.Logarithmic

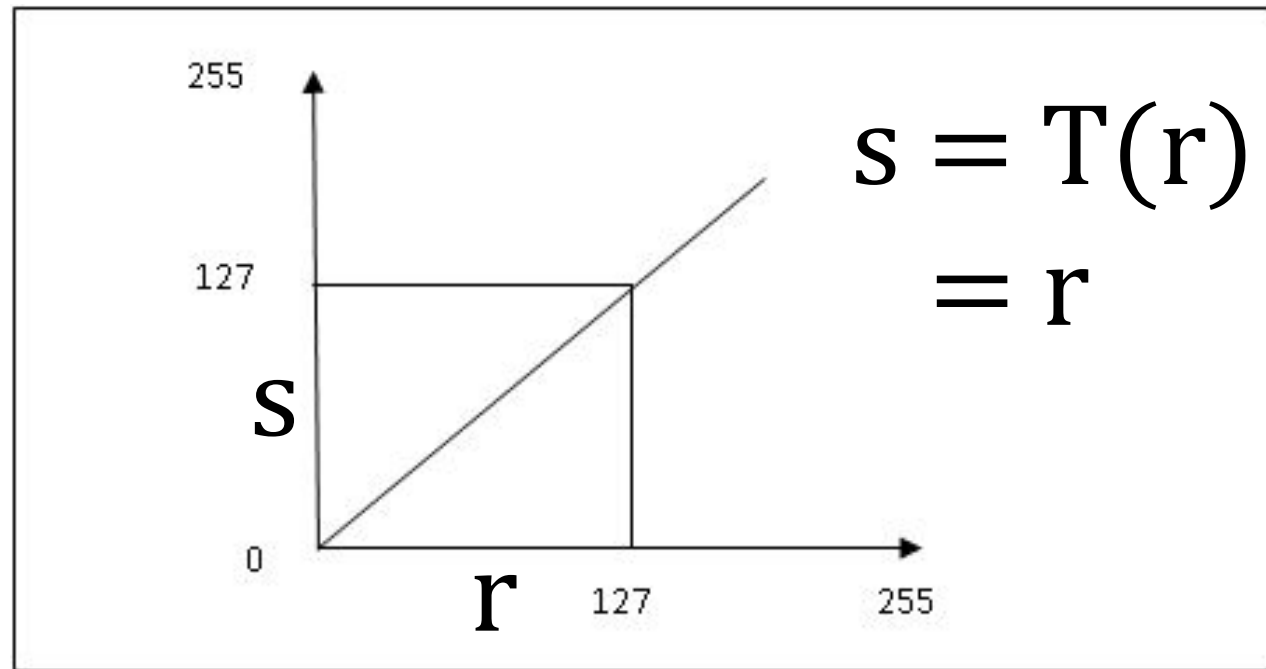
- 3.Power - law

Some basic Intensity transformation Functions

Linear

Identity

transformation : Each value of the image is directly mapped to each other values of the output image.



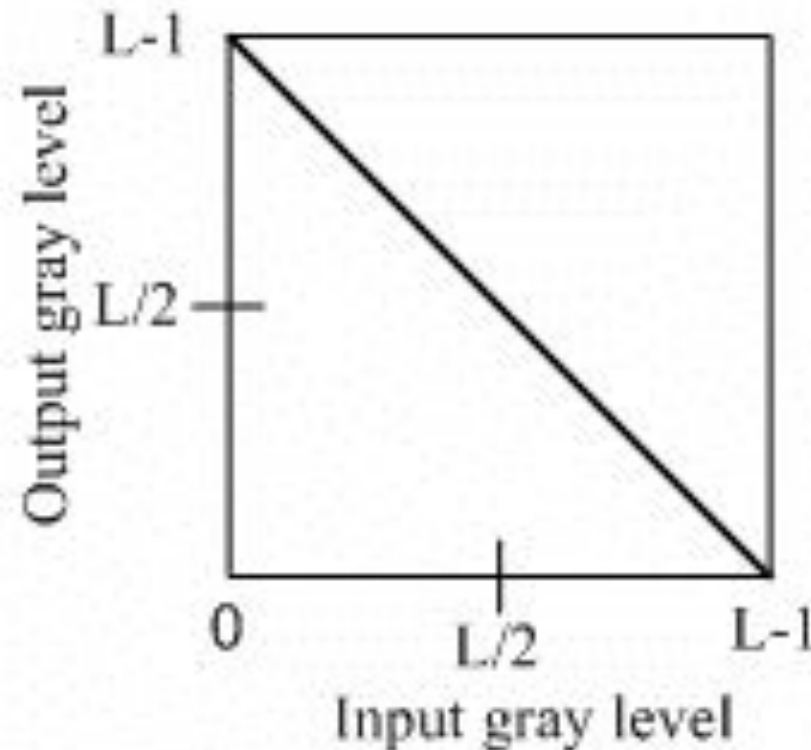
Identity transformation

Some basic Intensity transformation Functions

Linear

Negative

transformation : It is the opposite of identity transformation. Here, each value of the input image is subtracted from $L-1$ and then it is mapped onto the output image.

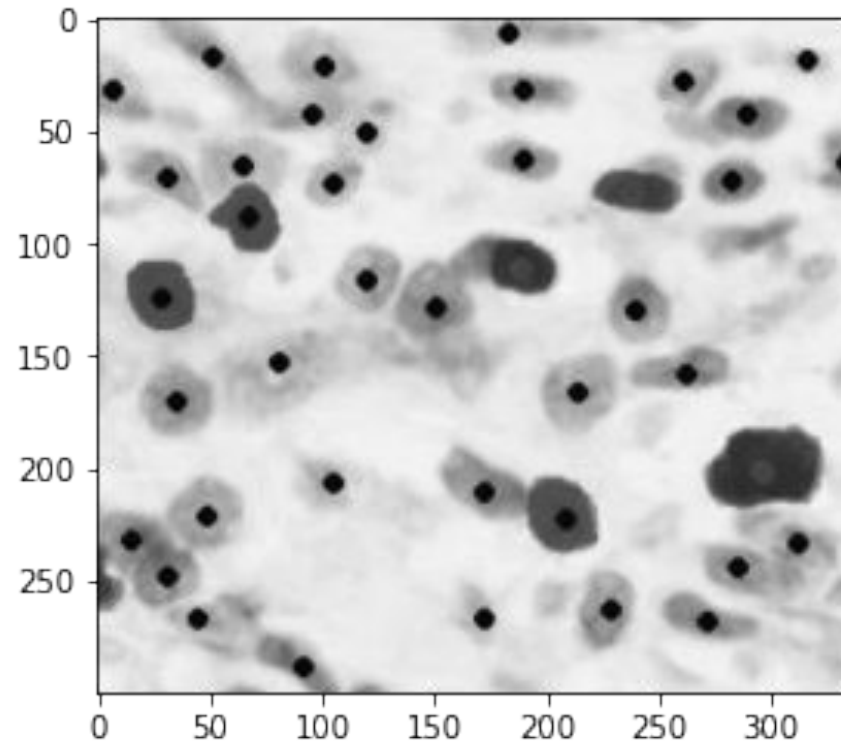
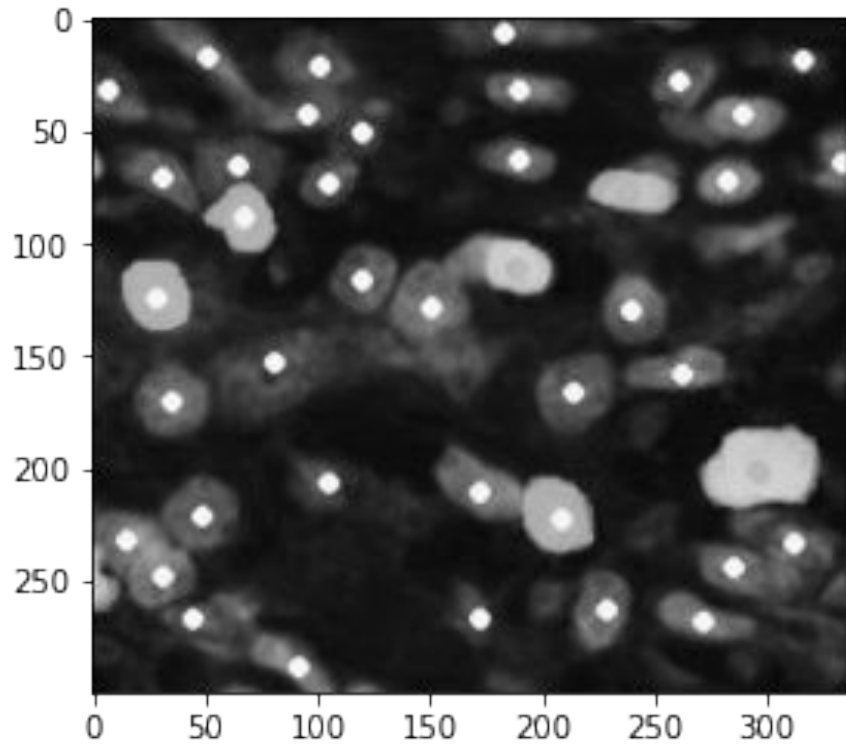


$$s = T(r) \\ = 255 - r$$

Some basic Intensity transformation Functions

Linear

Negative transformation



$$S = (L-1) - r$$

$$S = (8-1) - r$$

$$= 7 - r$$

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

Assuming the image to be 3-bit, we have $2^3 = 8$ levels.

$$r=0, S = 7-0 = 7$$

$$r=1, S = 7-1 = 6$$

$$r=2, S = 7-2 = 5$$

$$2^0 = 1$$

$$2^1 = 2$$

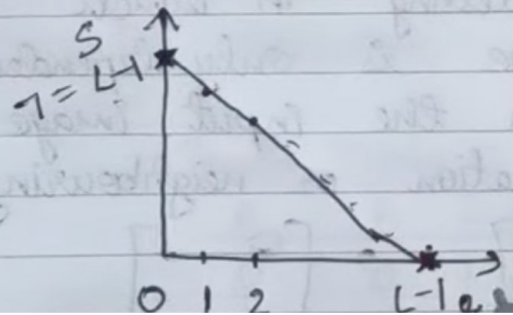
$$2^2 = 4$$

$$2^3 = 8 \quad 0-8$$

$$r=7, S = 7-7 = 0$$

$$[0, L-1] \quad n=3$$

$$L = 2^n = 2^3 = \underline{\underline{8}}$$



3	4	2	5
4	1	3	1
5	5	1	2
0	1	3	6

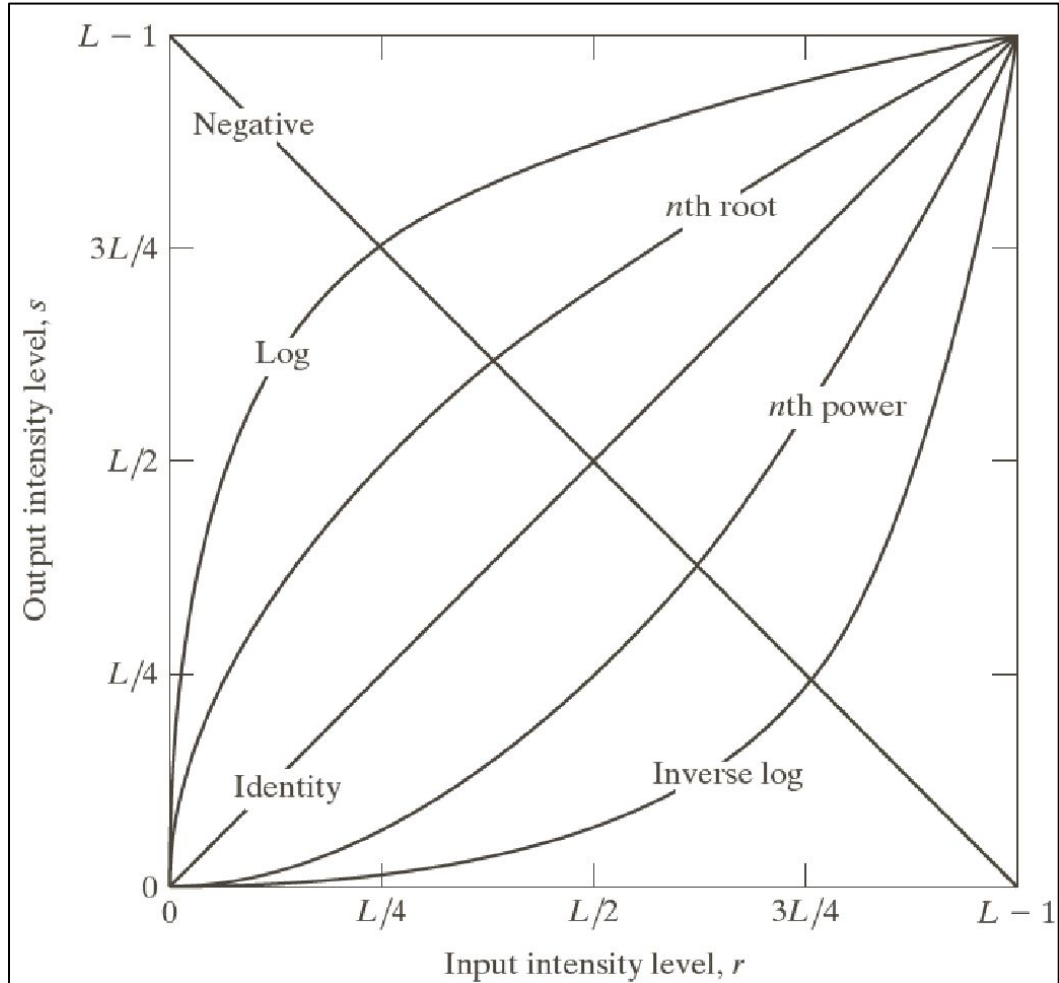
Digital Negative

Some basic Intensity transformation Functions

Logarithmic

Logarithmic transformation is divided into two types:

- ❑ Log transformation
- ❑ Inverse log transformation

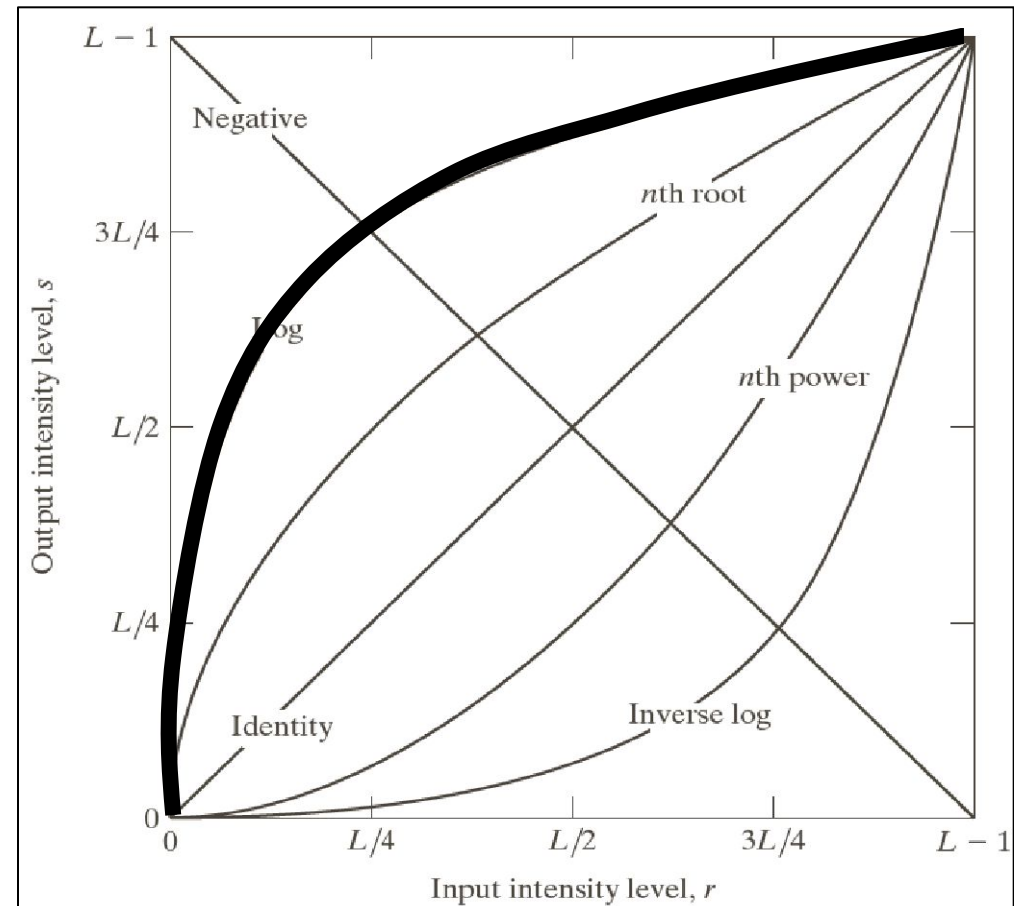


Some basic Intensity transformation Functions

Logarithmic

□ Log transformation

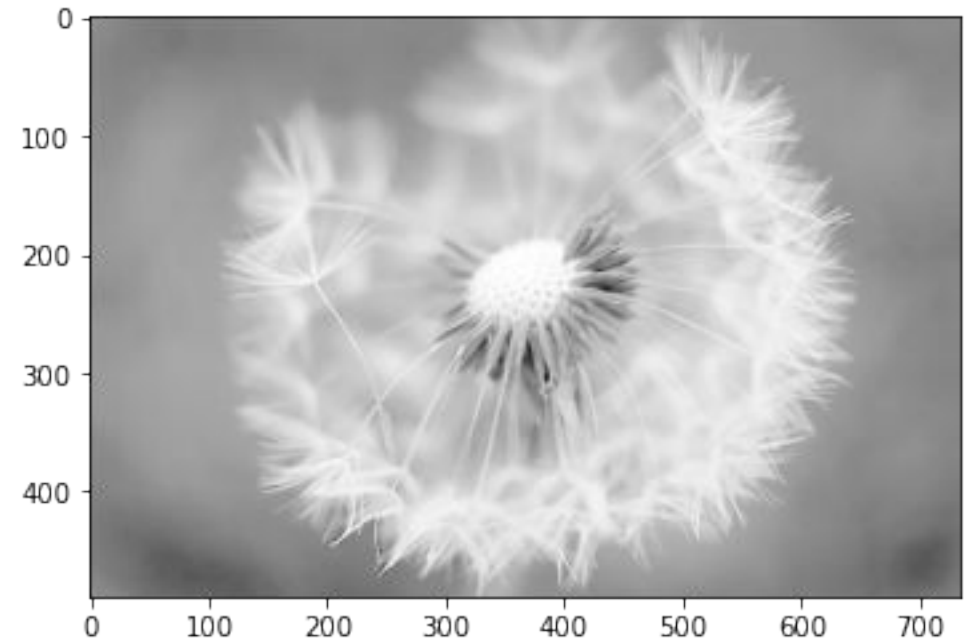
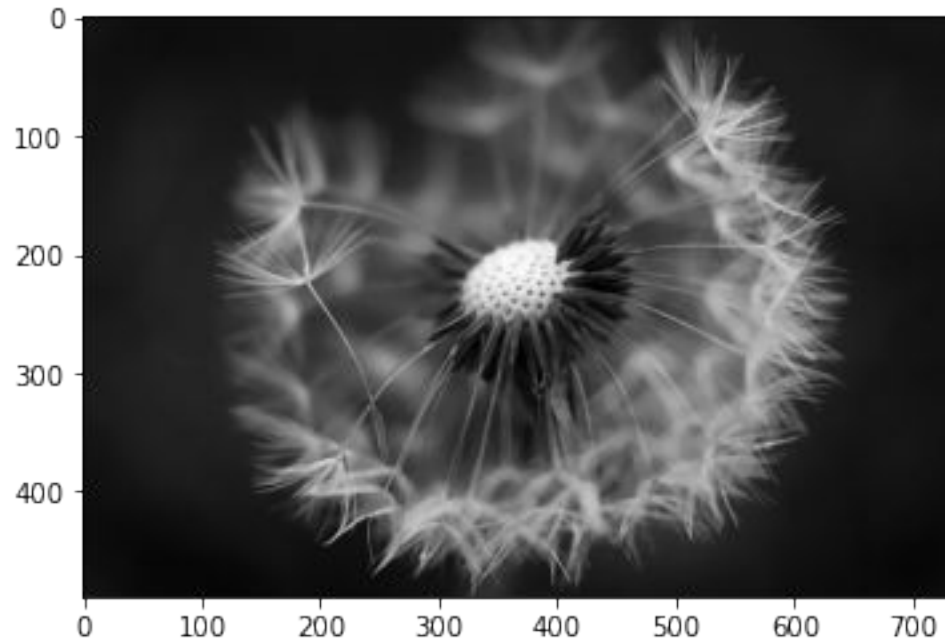
$$s = T(r) \\ = c * \log(1+r)$$



Some basic Intensity transformation Functions

Logarithmic

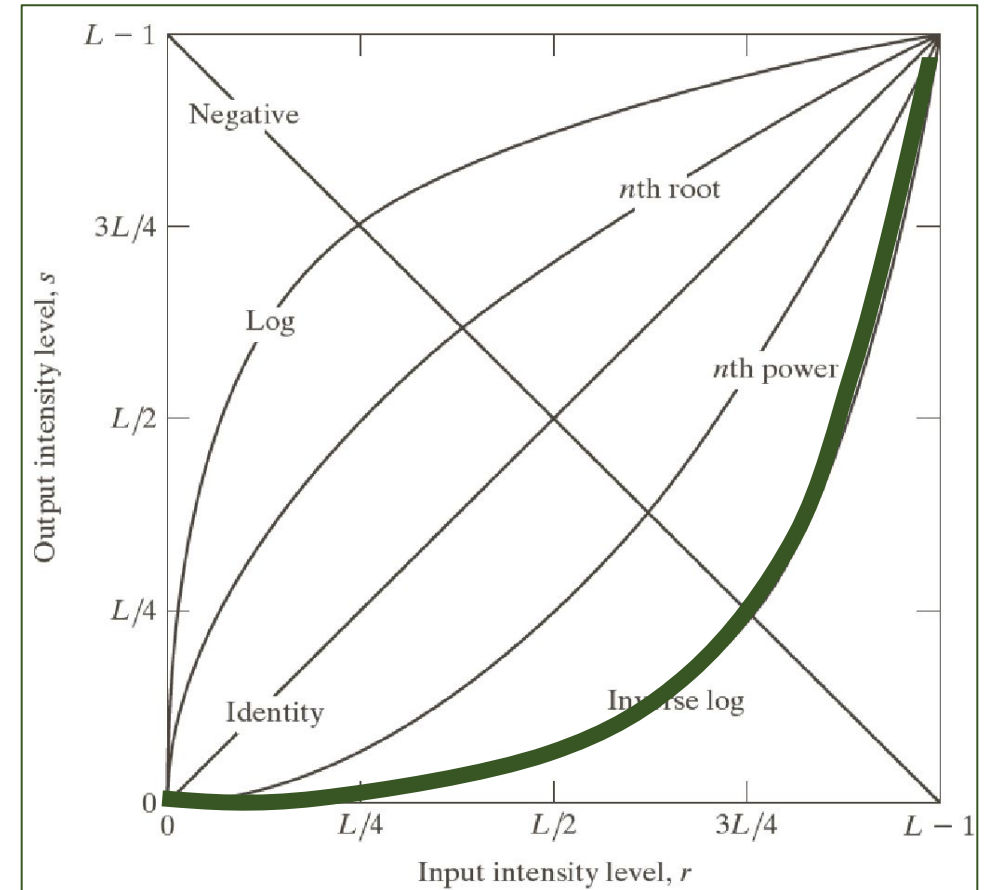
□ Log transformation



Some basic Intensity transformation Functions

Logarithmic

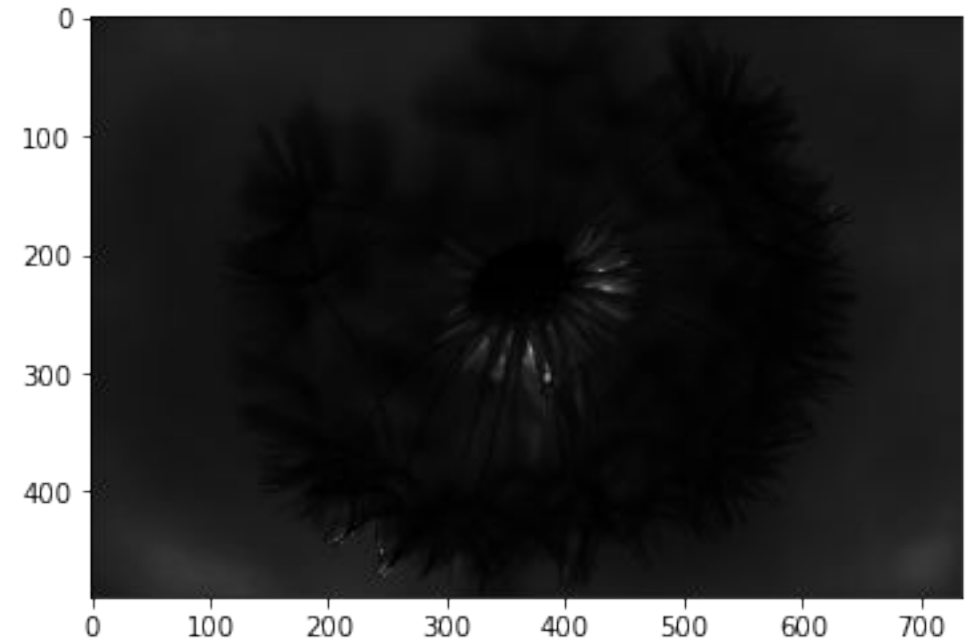
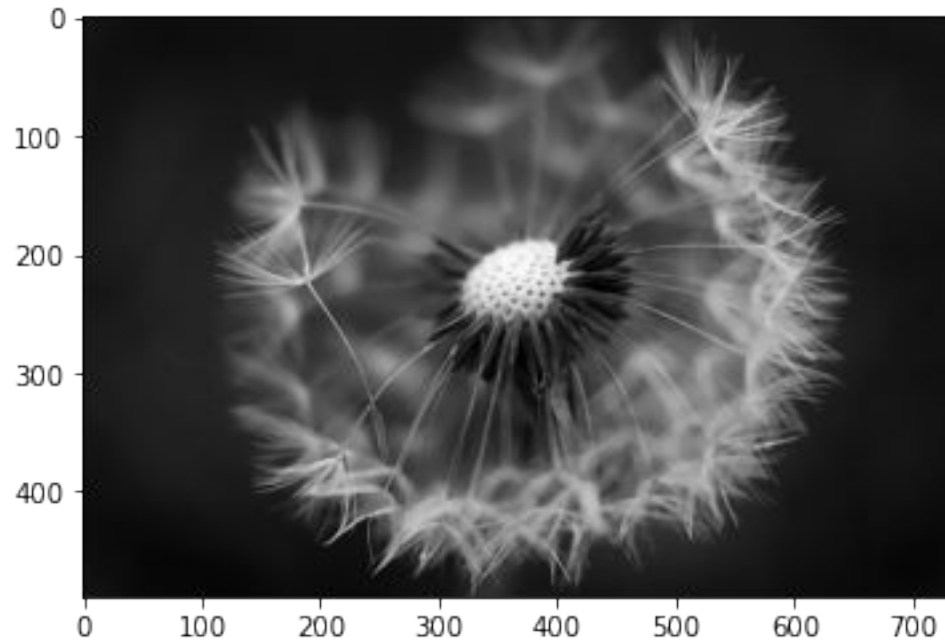
❑ Inverse log transformation



Some basic Intensity transformation Functions

Logarithmic

- ❑ Inverse Log transformation



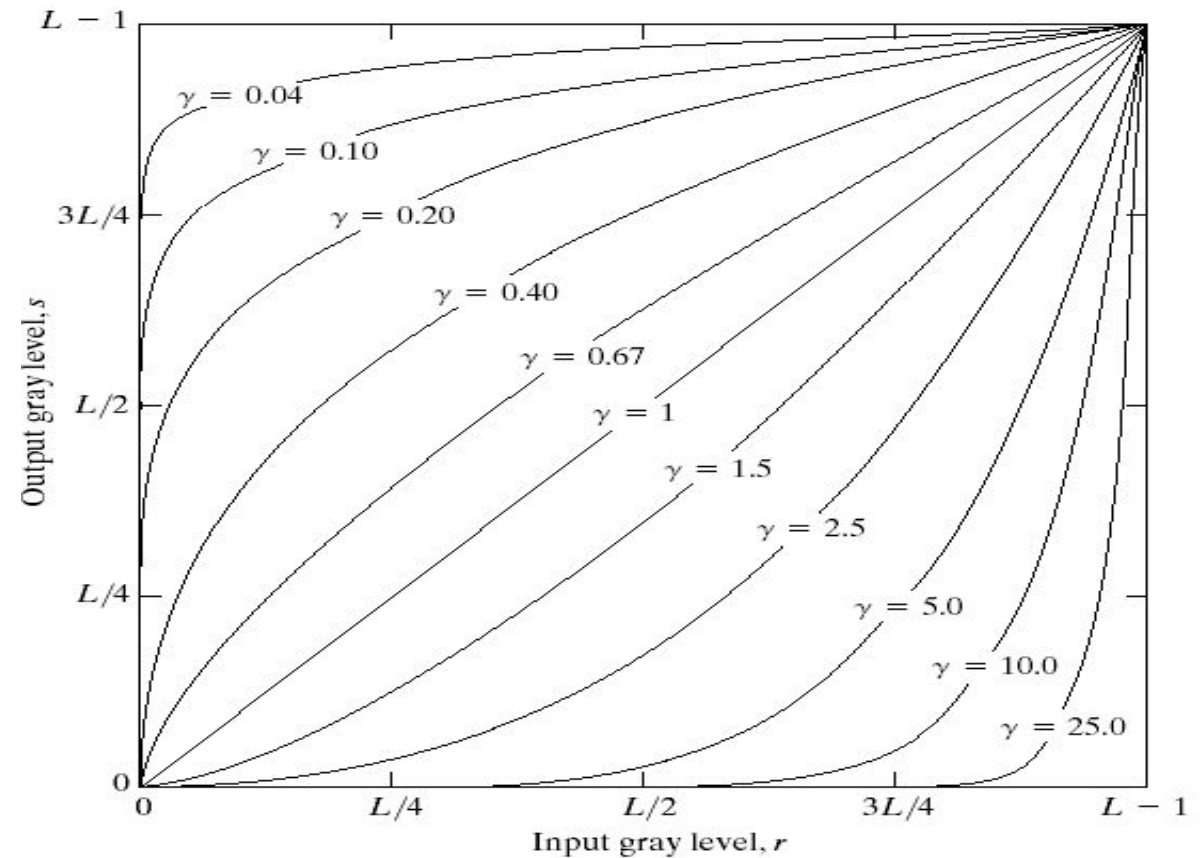
Power Law Transformations

Power law transformations have the following form

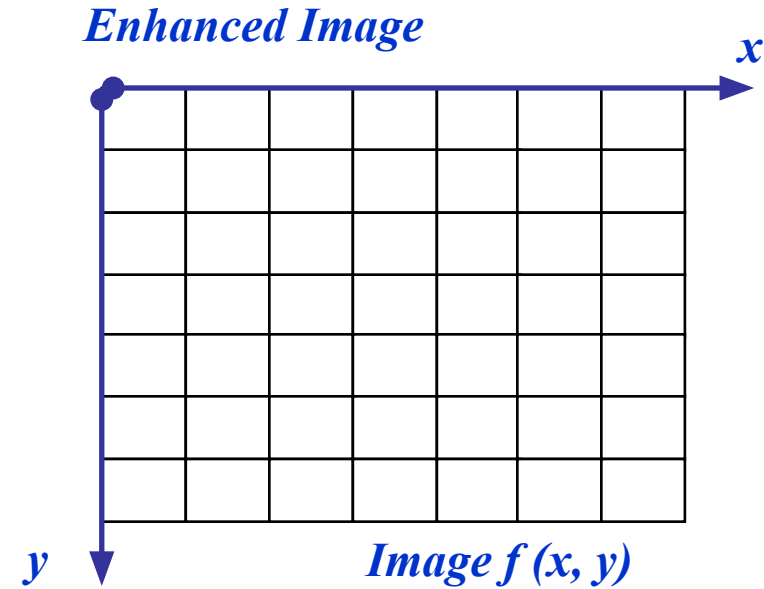
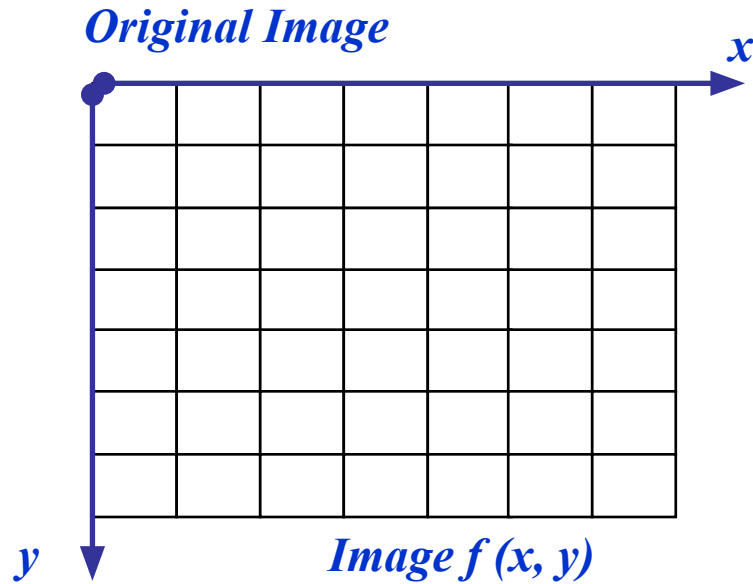
$$s = c * r^\gamma$$

Map a narrow range of dark input values into a wider range of output values or vice versa

Varying γ gives a whole family of curves



Power Law Transformations (cont...)



$$s = r^\gamma$$

We usually set c to 1

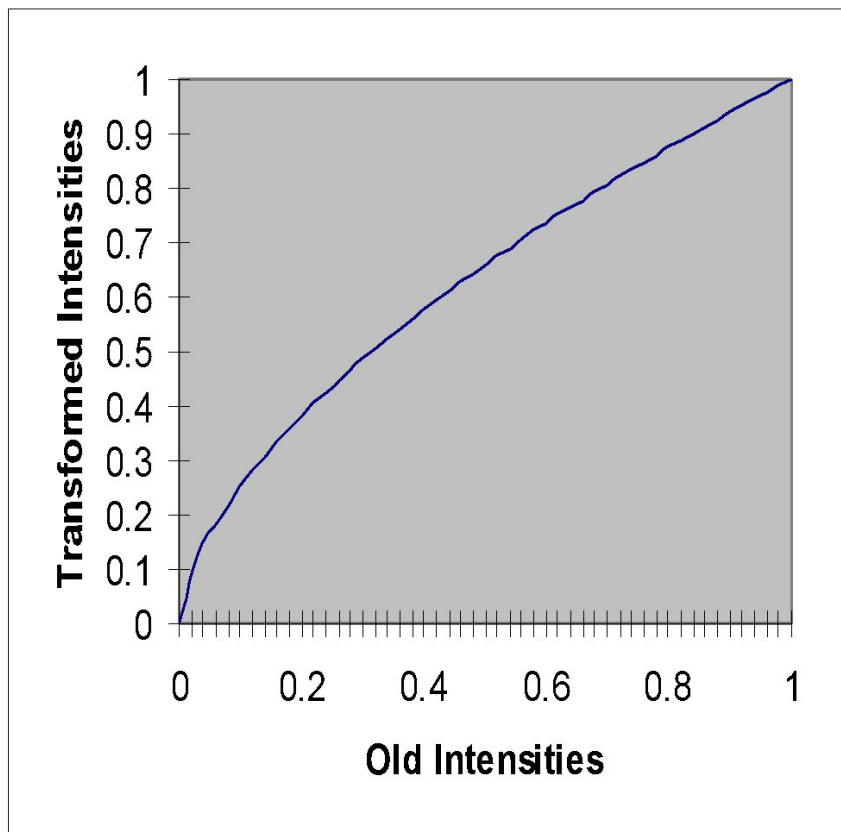
Grey levels must be in the range $[0.0, 1.0]$

Power Law Example



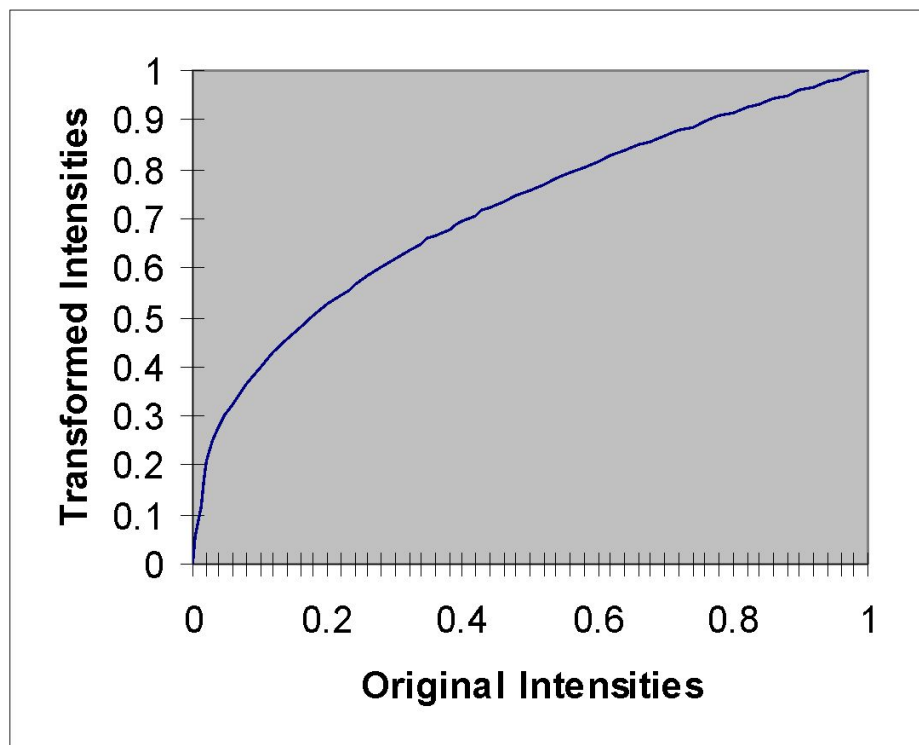
Power Law Example (cont...)

$$\gamma = 0.6$$



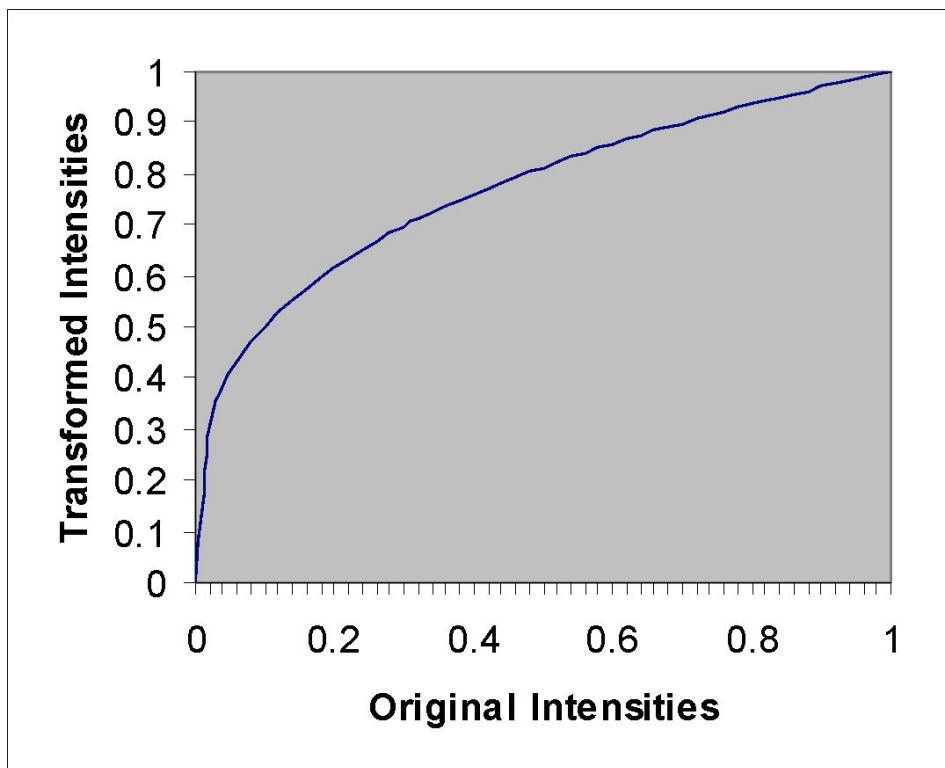
Power Law Example (cont...)

$$\gamma = 0.4$$



Power Law Example (cont...)

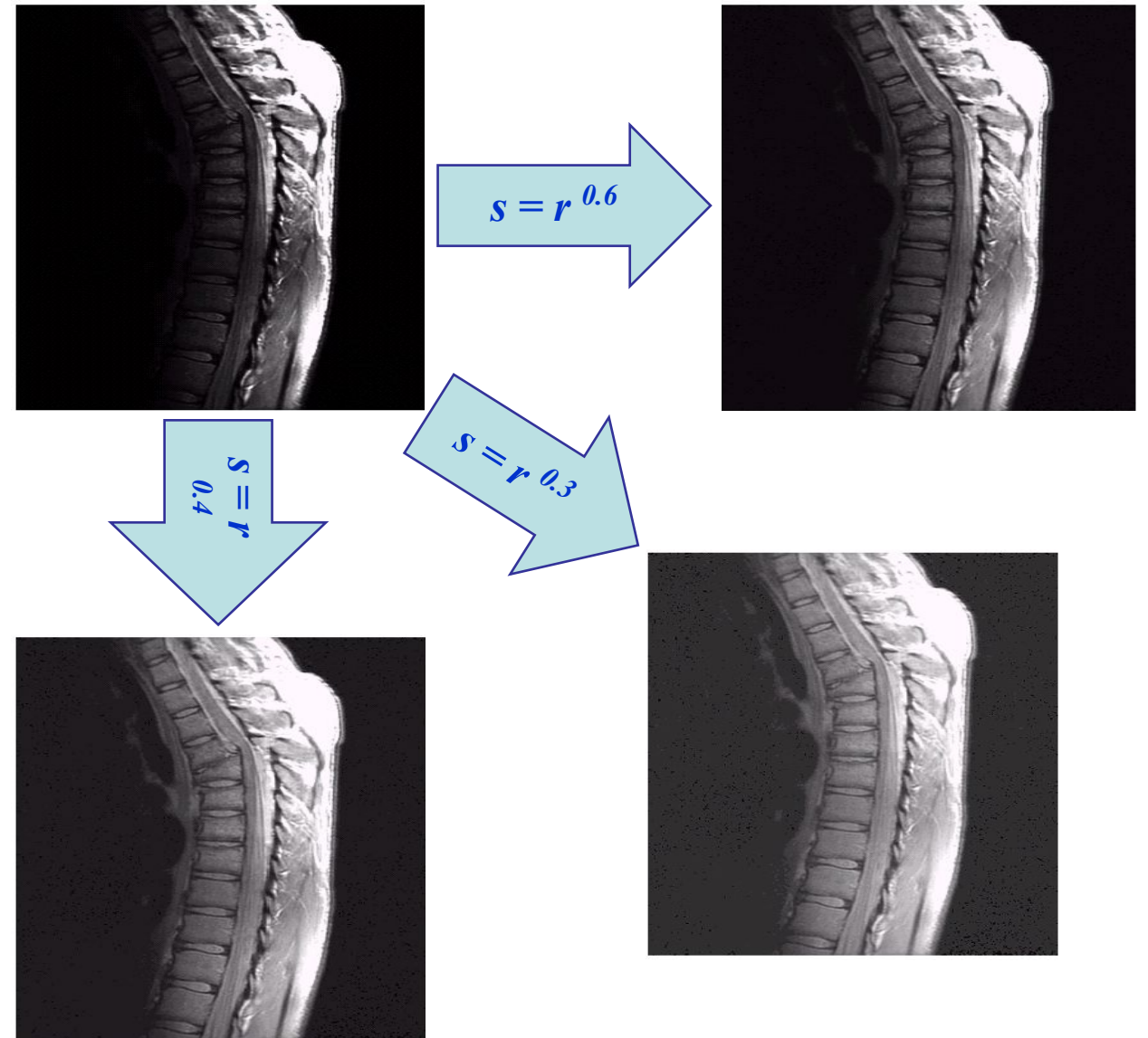
$$\gamma = 0.3$$



Power Law Example (cont...)

The images to the right show a magnetic resonance (MR) image of a fractured human spine

Different curves highlight different detail

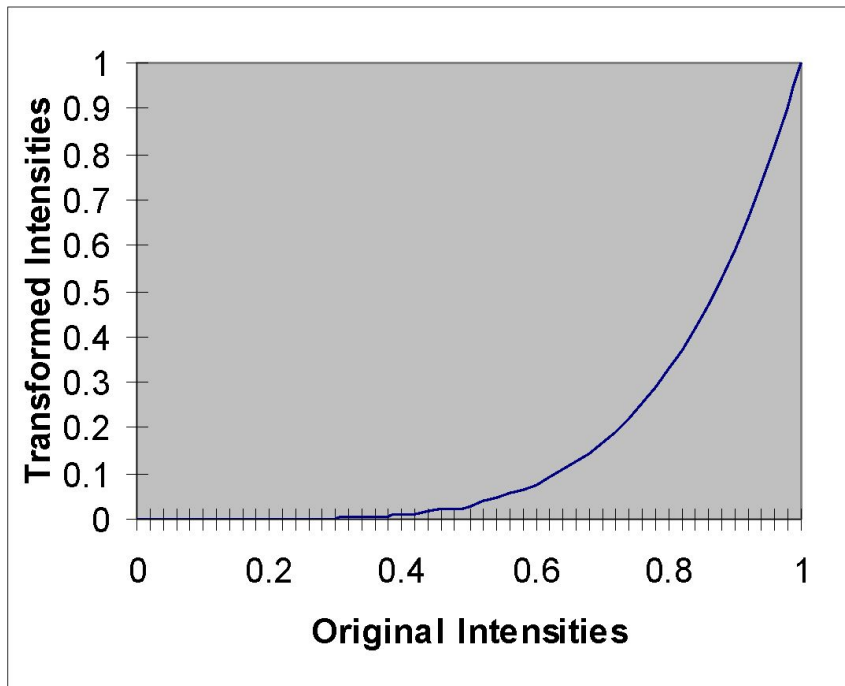


Power Law Example



Power Law Example (cont...)

$$\gamma = 5.0$$

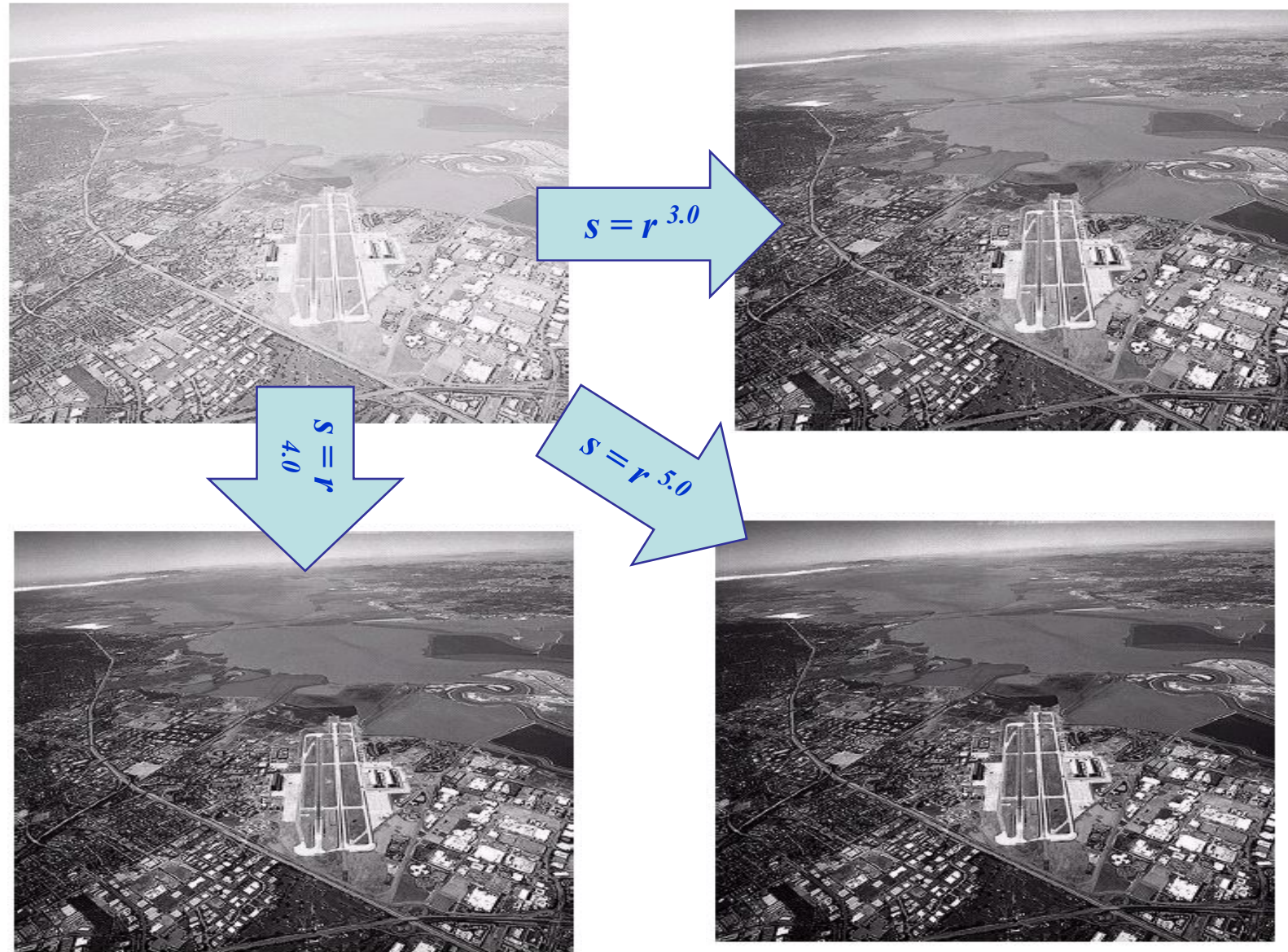


Power Law Transformations (cont...)

An aerial photo of a runway is shown

This time power law transforms are used to darken the image

Different curves highlight different detail



Some basic Intensity transformation Functions

Contrast stretching

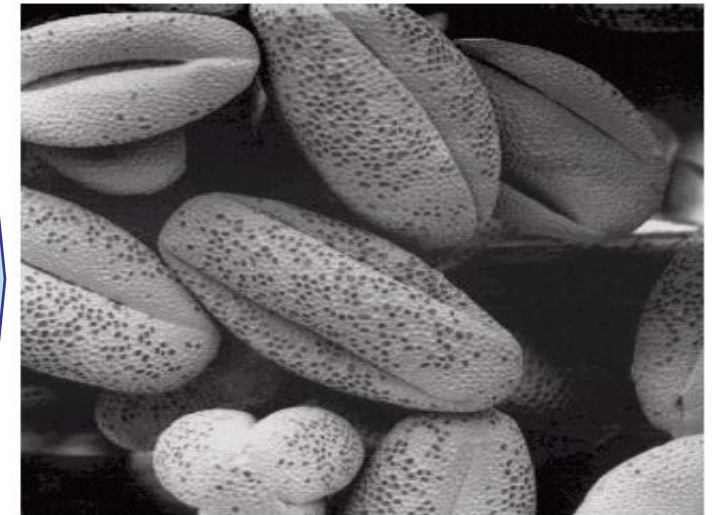
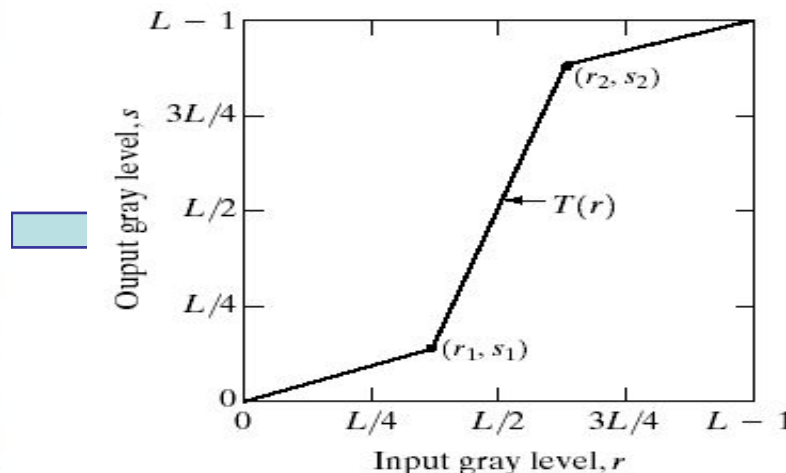
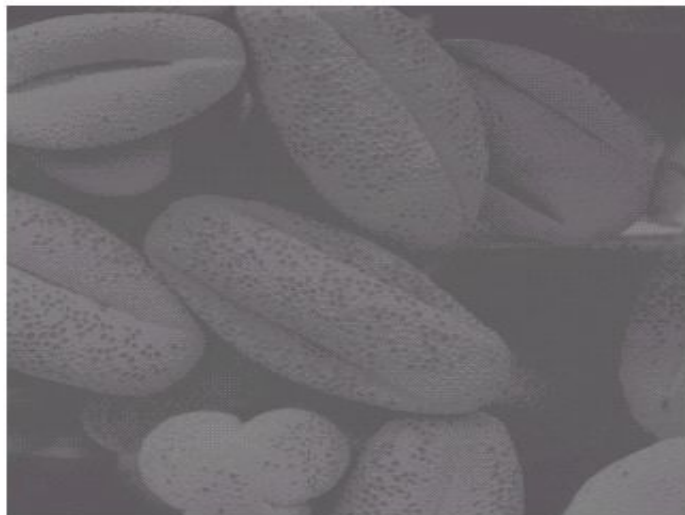
Contrast stretching (also called Normalization) attempts to improve an image by **stretching the range of intensity** values it contains to make full use of possible values.

Piecewise Linear Transformation Functions

Rather than using a well defined mathematical function we can use arbitrary user-defined transforms

The images below show a contrast stretching linear transform to add contrast to a poor quality image

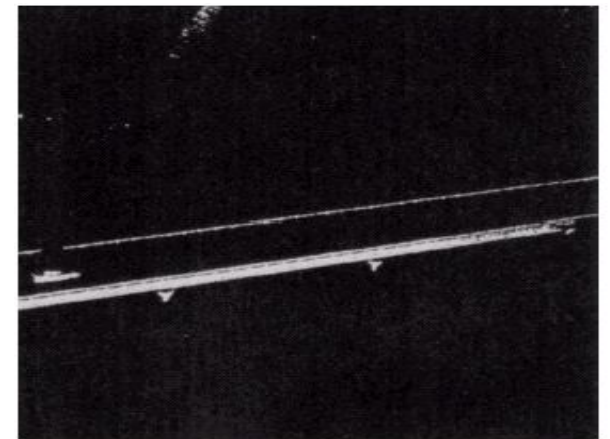
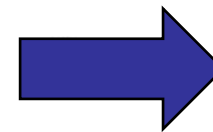
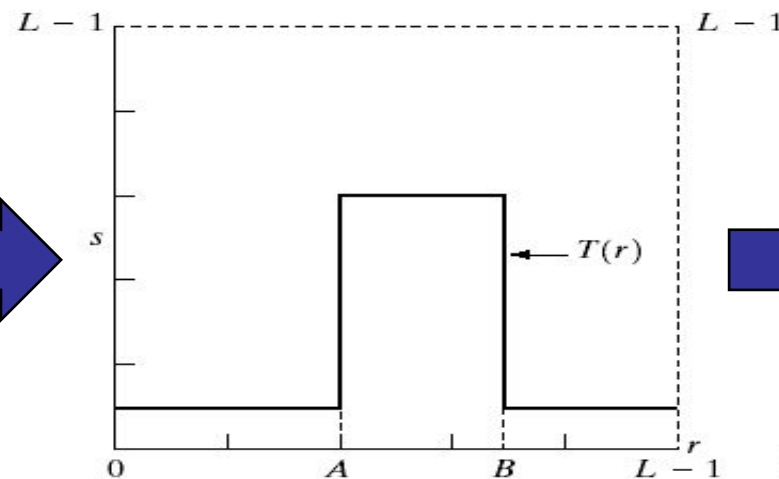
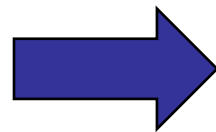
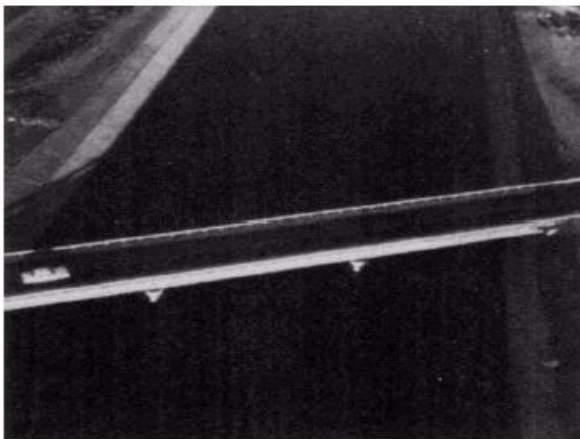
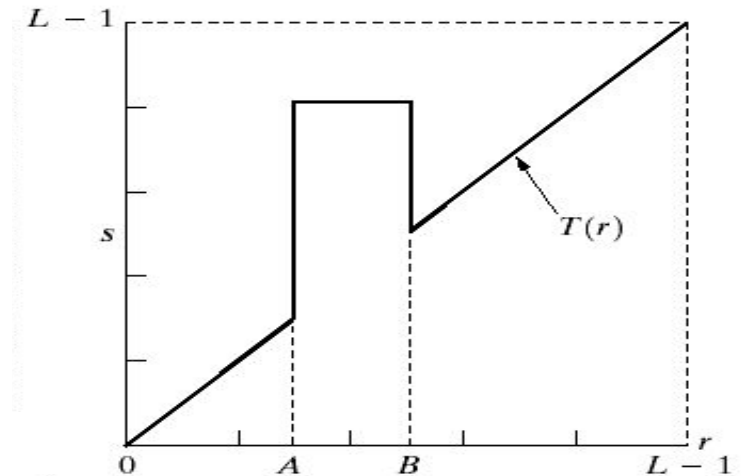
Contrast stretching (also called Normalization) attempts to improve an image by **stretching the range of intensity** values it contains to make full use of possible values.



Gray Level Slicing

Highlights a specific range of grey levels

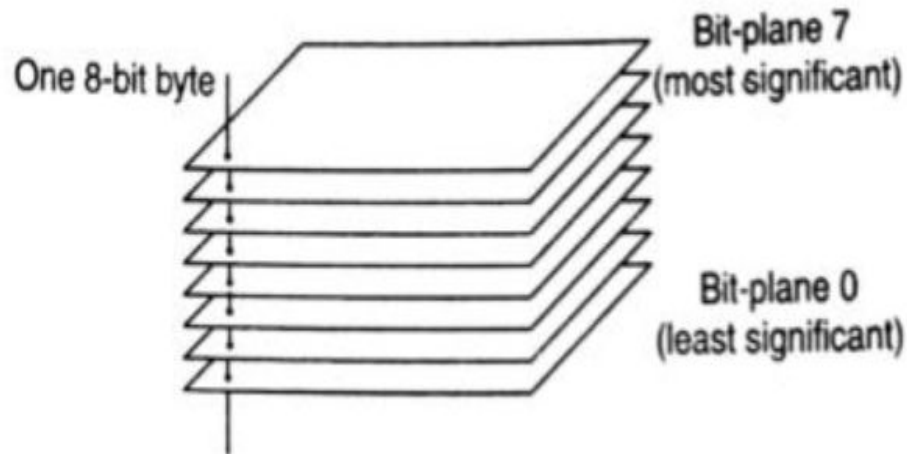
- Similar to thresholding
- Other levels can be suppressed or maintained
- Useful for highlighting features in an image



Some basic Intensity transformation Functions

Bit plane slicing

Bit plane slicing is a method of representing an image with one or more bits of the byte used for each pixel.



One can use only MSB to represent the pixel, which reduces the original gray level to a binary image. The three main goals of bit plane slicing is:

- ☐ Converting a gray level image to a binary image.
- ☐ Representing an image with fewer bits
- ☐ Enhancing the image by focussing.

Some basic Intensity transformation Functions

Bit plane slicing

6	7	6	6	7
0	0	0	1	2
1	1	1	2	3
4	5	5	4	2
6	6	6	7	7

110	111	110	110	111
000	000	000	001	010
001	001	001	010	011
100	101	101	100	010
110	110	110	111	111

1	1	1	1	1
0	0	0	0	0
0	0	0	0	0
1	1	1	1	0
1	1	1	1	1

MSB plane

1	1	1	1	1
0	0	0	0	1
0	0	0	1	1
0	0	0	0	1
1	1	1	1	1

Centre bit plane

0	1	0	0	1
0	0	0	1	0
1	1	1	0	1
0	1	1	0	0
0	0	0	1	1

LSB plane

Bit plane slicing

An 8-bit grayscale image



Bitplane #8



Bitplane #7



Bitplane #6



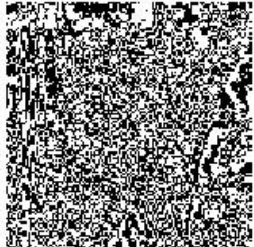
Bitplane #5



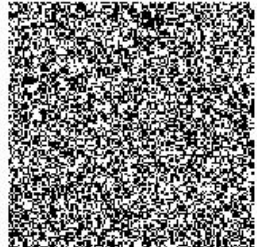
Bitplane #4



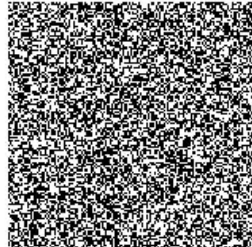
Bitplane #3



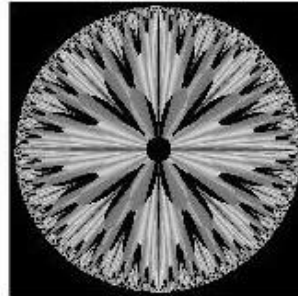
Bitplane #2



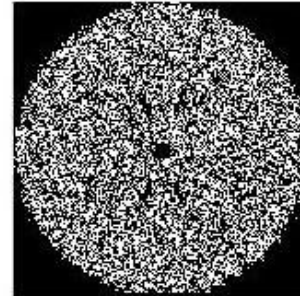
Bitplane #1



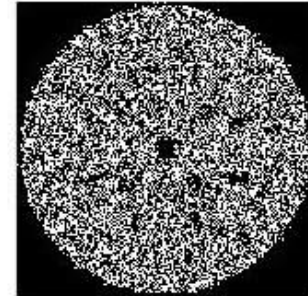
Original Image



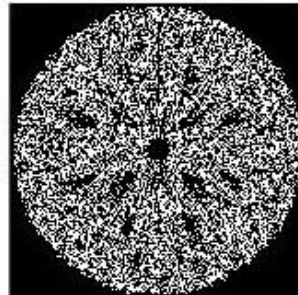
Bit plane 1



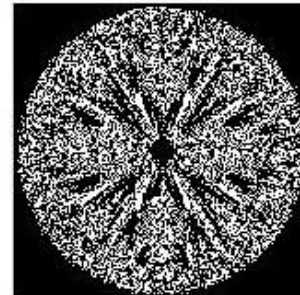
Bit plane 2



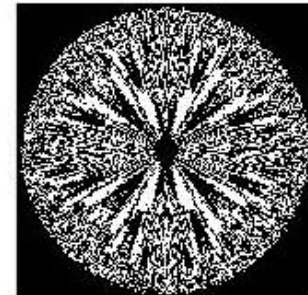
Bit plane 3



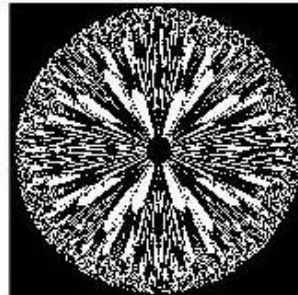
Bit plane 4



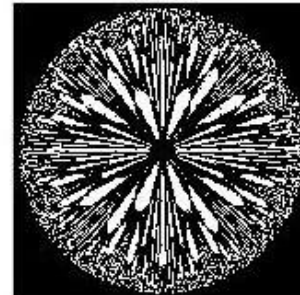
Bit plane 5



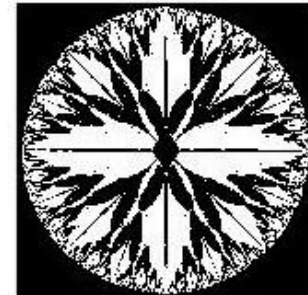
Bit plane 6



Bit plane 7



Bit plane 8



Some basic Intensity transformation Functions

Bit Plane Slicing



Histogram equalization

Histogram

Histogram is a graphical representation of the **intensity distribution** of an image. In simple terms, it represents the **number of pixels** for **each intensity** value considered.

Histogram equalization

Histogram Equalization: It is a computer image processing technique used to improve **contrast in images** . It accomplishes this by effectively **spreading out the most frequent intensity** values, i.e. **stretching out the intensity range** of the image. This method usually increases the **global contrast** of images when its usable data is represented by **close contrast values** . This allows for areas of lower **local contrast** to gain a higher contrast.

Histogram Equalisation

Spreading out the frequencies in an image (or equalising the image) is a simple way to improve dark or washed out images

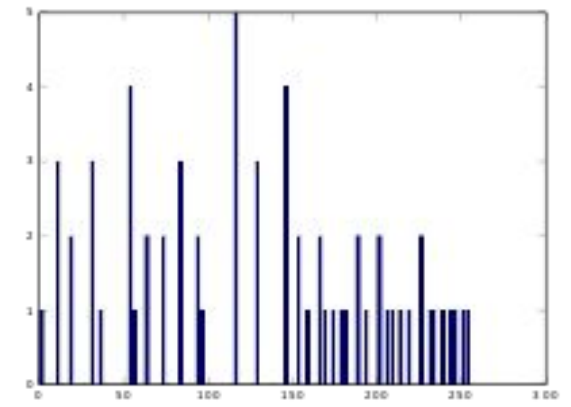
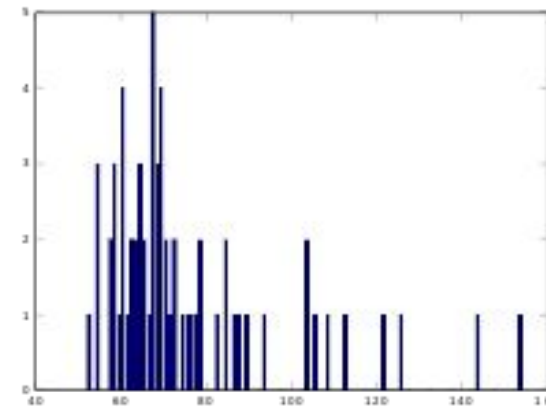
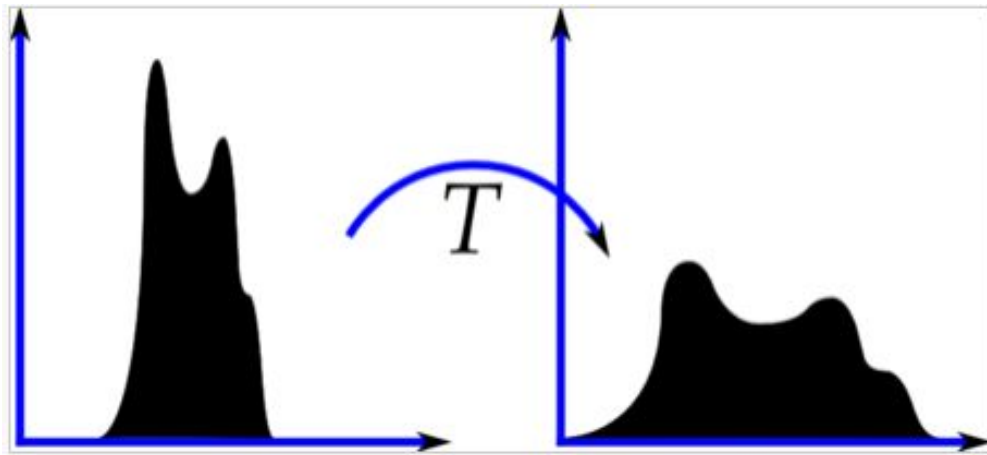
The formula for histogram equalisation is given where

- r_k : input intensity
- s_k : processed intensity
- k : the intensity range
(e.g 0.0 – 1.0)
- n_j : the frequency of intensity j
- n : the sum of all frequencies

$$\begin{aligned} s_k &= T(r_k) \\ &= \sum_{j=1}^k p_r(r_j) \\ &= \sum_{j=1}^k \frac{n_j}{n} \end{aligned}$$

Histogram equalization

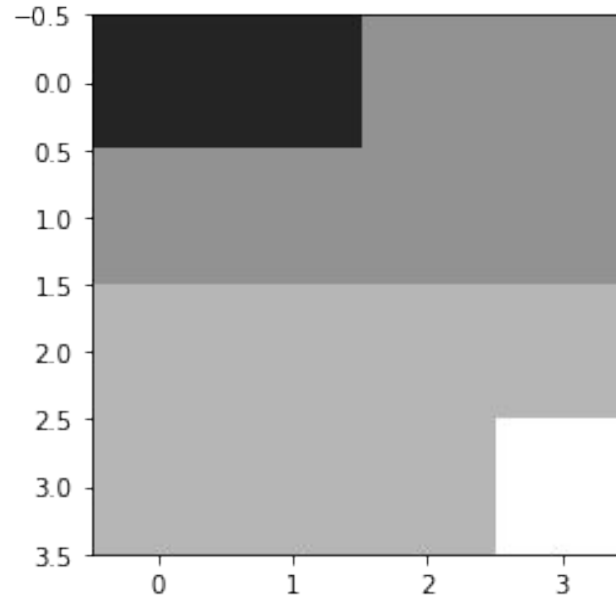
The Histogram Equalization algorithm enhances the contrast of images by transforming the values in an intensity image so that the histogram of the output image is approximately flat.



Histogram equalization

Count the frequency

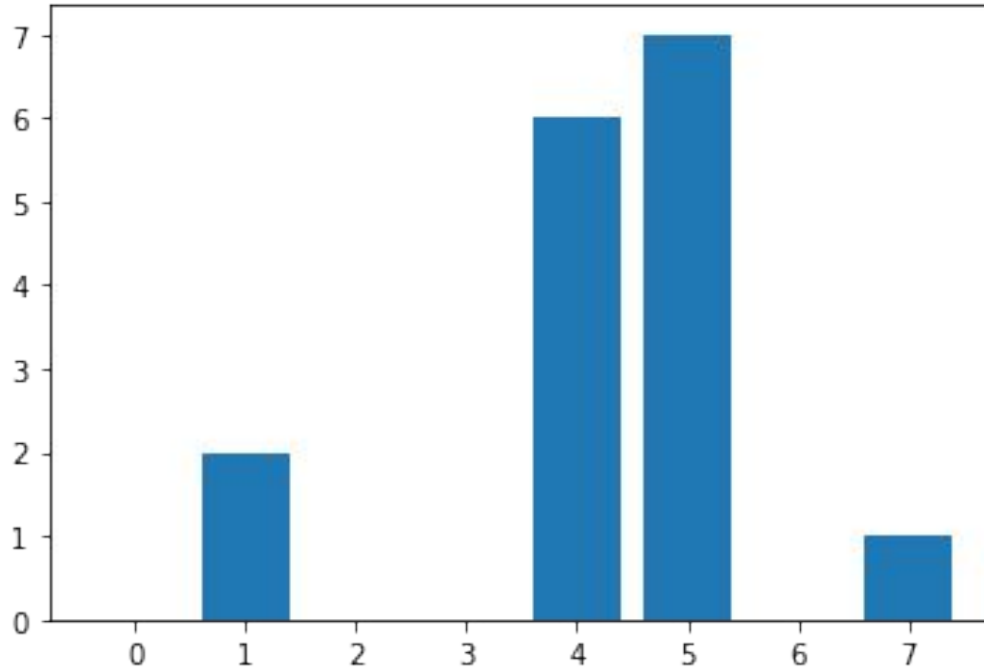
1	1	4	4
4	4	4	4
5	5	5	5
5	5	5	7



Intensity	Freq
0	0
1	2
2	0
3	0
4	6
5	7
6	0
7	1

Histogram equalization

Draw the histogram



Intensity	Freq
0	0
1	2
2	0
3	0
4	6
5	7
6	0
7	1

Histogram equalization

The Probability Density Function(PDF) defines the probability function representing the density of a continuous random variable lying between a specific range of values.

PMF - Probability mass function refers to discrete probabilities.

Histogram equalization

Calculate PMF for our intensities

Intensity	Freq	PMF (Freq/Total Pixels)
0	0	0
1	2	0.125
2	0	0
3	0	0
4	6	0.375
5	7	0.4375
6	0	0
7	1	0.0625

Histogram equalization

The cumulative distribution function (**CDF**) calculates the cumulative probability for a given x-value. It can be used as mapping function for equalization.

It is statistically proven that if we use CDF as Transformation function the PDF of the output image becomes uniform.

Histogram equalization

Calculating CDF

Intensity	Freq	PMF (Freq/Total Pixels)	CDF
0	0	0	0
1	2	0.125	0.125
2	0	0	0.125
3	0	0	0.125
4	6	0.375	0.5
5	7	0.4375	0.9375
6	0	0	0.9375
7	1	0.0625	1

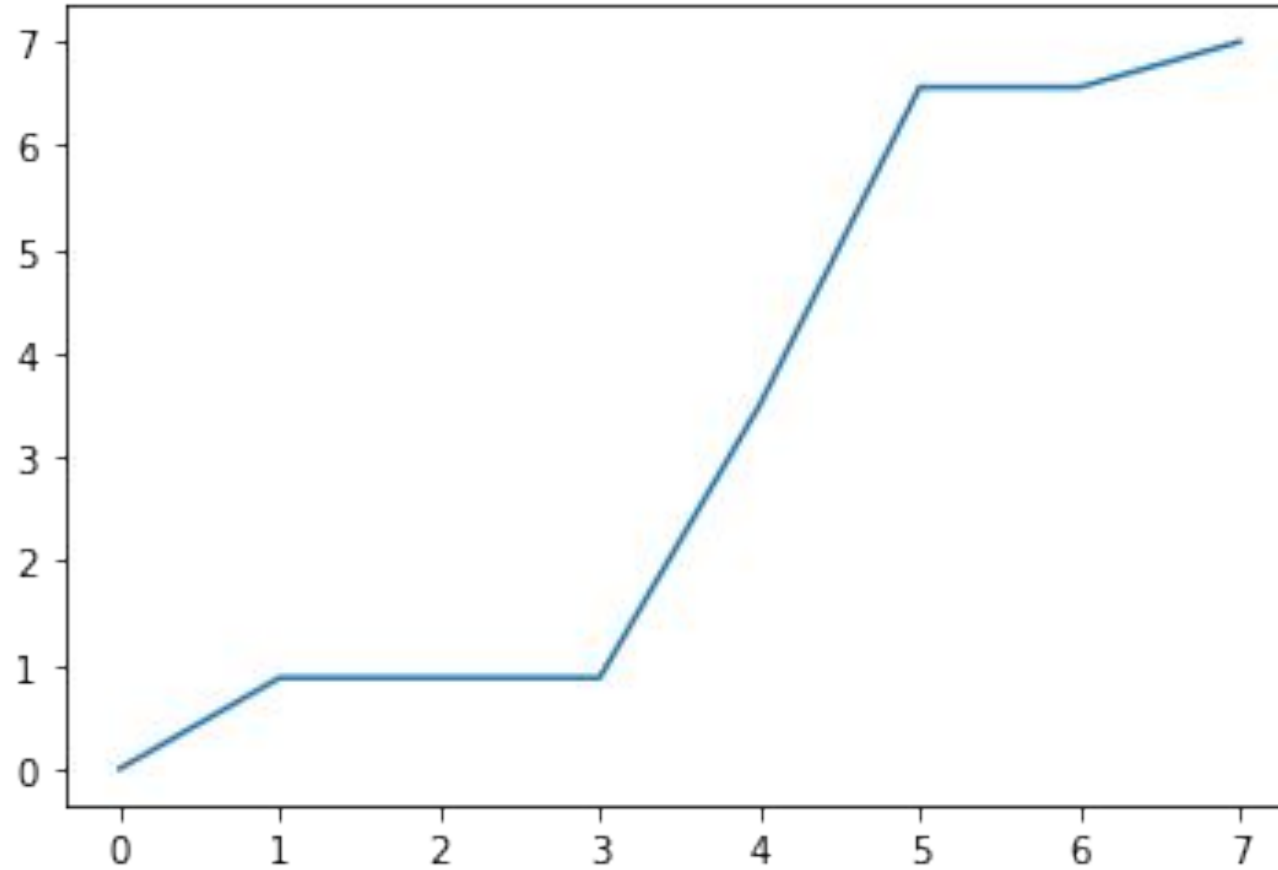
Histogram equalization

Converting CDF to required gray range

Intensity	Freq	PMF (Freq/Total Pixels)	CDF	CDF*(L-1)
0	0	0	0	0
1	2	0.125	0.125	0.875
2	0	0	0.125	0.875
3	0	0	0.125	0.875
4	6	0.375	0.5	3.5
5	7	0.4375	0.9375	6.5625
6	0	0	0.9375	6.5625
7	1	0.0625	1	7

Histogram equalization

CDF Function



Histogram equalization

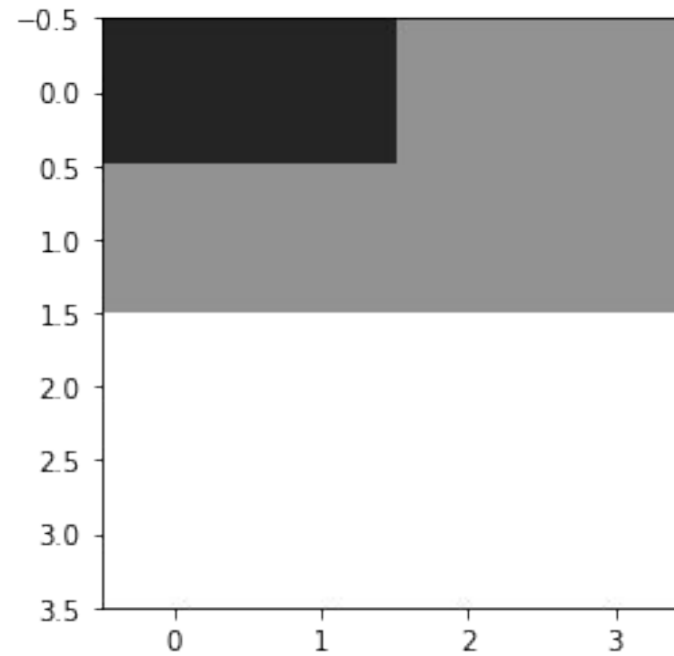
Transforming levels to nearest integers

Intensity	Freq	PMF (Freq/Total Pixels)	CDF	CDF*(L-1)	Output Intensity level
0	0	0	0	0	0
1	2	0.125	0.125	0.875	1
2	0	0	0.125	0.875	1
3	0	0	0.125	0.875	1
4	6	0.375	0.5	3.5	4
5	7	0.4375	0.9375	6.5625	7
6	0	0	0.9375	6.5625	7
7	1	0.0625	1	7	7

Histogram equalization

New Image

1	1	4	4
4	4	4	4
7	7	7	7
7	7	7	7



Histogram equalization

Histogram Comparison

